

Perceptual integrality does not entail language-wide enhancement of contrasts

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Abstract

Perceptual integrality is found when perceivers do not judge differences on physically orthogonal dimensions independently (Garner, 1974). For example, F1 and nasality in vowels do not necessarily physically covary, but native English listeners judge them as if they covary negatively (Kingston & Macmillan, 1995). Kingston and Diehl (1994) posited that if integrality of two phonetic properties increases perceptual distance between categories differing in both, language users covary those properties in production, enhancing the contrast. This study investigates whether native English speakers exploit integrality of F1 and nasality to enhance vowel height contrasts by covarying them in production.

Perceptual and production results paired within participants showed that F1-nasality covariation in production was predictable partly from how perceptually integrated F1 and nasality were and partly from lexical factors such as the presence of minimal pairs depending on vowel height. These results are inconsistent with enhancement of language-wide phonological contrasts, but compatible with enhancement of contrasts between lexical items.

1 Introduction

A persistent question in phonetics is whether the perceptual utilization of phonetic properties for categorization predicts how those properties are manipulated in production. Kingston and Diehl (1994) propose that if two phonetic properties are perceptually *integral* (perceived non-independently), speakers may covary those properties in their productions to take advantage of this non-independent perception and increase the perceptual distance between categories differing in both properties.

Perceptual integrality is said to occur when perceivers do not judge separate properties of a stimulus independently even when they are physically orthogonal (Garner, 1974). For example, vowels may vary independently in F1 (height) and the intensity of nasal poles and zeroes (nasality), but native listeners of English do not perceive these two phonetic properties independently. Rather, they treat pairs of vowels in which F1 and nasality are negatively correlated as more perceptually distinct than pairs of vowels in which they are positively correlated, i.e., the difference between high and nasal vowels vs. non-high and oral vowels was greater than high and oral vs. non-high and nasal (Kingston & Macmillan, 1995; Macmillan et al., 1999). This can be understood as the (partial) collapse of multiple physical dimensions to a single perceptual dimension. If changes on two different physical dimensions produce similar perceptual effects, these coinciding perceived changes may add together to increase perceptual distance. For example, if F1 and the nasal pole are both low, they may produce the same perceptual effect of highly concentrated low-frequency energy in a vowel's spectrum which is interpreted as vowel raising; if they are both high, then they may produce the perceptual effect of vowel lowering in the same manner. These two percepts will be more distinct from each other than the percepts that might arise from F1 being low and the nasal pole being high or

vice versa, in which case the combined perceptual effect will be an absence of such highly concentrated energy anywhere in the vowel’s spectrum.

One possible consequence of integrality is that when two phonetic properties A and B are perceptually integral and there are two or more categories which differ in both A and B, language users may covary A and B in production to exploit their integrality and *enhance* the contrast between categories in perception (the ‘enhancement hypothesis’, Kingston & Diehl, 1994). Continuing with F1 and nasality as the example, if these two properties are perceived as covarying negatively (i.e., higher vowels sound more nasal) then English speakers might turn the perceived covariation into actual covariation in their productions, enhancing height contrasts by modulating nasality or (because English listeners use a vowel’s nasality as evidence for a word’s identity even if nasality is not phonologically contrastive in vowels; Beddor et al., 2018; Ohala and Ohala, 1995) enhancing nasality contrasts by modulating F1.

The enhancement hypothesis at its most basic states that language users may covary two phonetic properties in their productions so as to exploit perceptual integrality between them and increase the perceptual distance between categories that differ in those properties. In addition to the obvious task of deciding on a pair of phonetic properties to focus on, an investigation of the enhancement hypothesis must also answer two crucial questions: ‘How is enhancement distributed between or within language users?’ and ‘Between which categories are contrasts enhanced?’

In answering the first question, this study has attempted to follow the recommendations of Schertz and Clare (2020), who conduct an extensive review of studies of the production-perception interface and identify as a dimension of variation the level of abstraction/granularity at which these studies test their hypotheses. This continuum ranges from tests of production-perception corre-

lations at the level of whole speech communities to investigations of variation in production-perception correlations within individuals. Among other recommendations, Schertz and Clare argue that researchers should explicitly situate their investigations of the production-perception interface on this continuum and that doing so will increase comparability between studies. Accordingly, this study attempts to explicitly convey the level of abstraction at which each part of the analysis has been conducted, as well as the degree to which high-level summaries are representative of lower-level variation.

Answering the second question, ‘Between which categories are contrasts enhanced?’, requires that Schertz and Clare’s continuum of abstraction between group-level and within-individual hypothesis testing be expanded to a space with a second dimension of variation: the level of abstraction at which contrasting categories are defined, and relatedly, the factors which can affect categorization decisions. Schertz and Clare implicitly frame contrast and categorization in terms of phonological categories that contrast at a language-wide level, i.e., phonemes and/or phonological features; many of the studies they reference do so explicitly, as do Kingston and Diehl (1994). However, there is ample evidence that speech perception involves the continuous categorization of auditory input not (only) as phonemes but as words (Holt & Lotto, 2010), and that these categorization decisions are affected by factors at the segmental and lexical levels, among others (Cutler, 2012, ch. 11). Minimally, investigations of the production-perception interface should explicitly specify the level of abstraction at which they test their hypotheses not only in terms of production-perception correlations at the group, individual, or within-individual level, but also in terms of production-perception correlations within different notions of what constitutes a contrasting category such as features, phonemes, words, etc.

Returning to the definition of the enhancement hypothesis given earlier,

the question that this study attempts to answer may be stated as ‘Do native speakers of English covary F1 and nasality in their productions so as to exploit perceptual integrality between them and increase the perceptual distance between phonemes and/or words which differ in those properties?’ To answer this question, paired production and perception data were gathered to assess within-participant and group-level correlations between F1-nasality covariation in production and F1-nasality integrality in perception, directly testing the two simplest versions of the enhancement hypothesis: enhancement consistent across a group, and enhancement consistent within individuals. If enhancement occurs, the degree to which F1 and nasality are integral in perception should predict the degree to which they covary in production.

In Experiment 1, participants read a word list while oronasal airflow measurements were taken. The vowel in the primary stressed syllable of each word varied in i) its height and ii) the orality or nasality of the immediately following consonant. In Experiment 2, the degree of perceptual integrality of F1 and nasality was measured using the Garner paradigm (Garner, 1974). This paradigm was employed in previous investigations of integrality between F1 and nasality which have demonstrated its existence in native English listeners (Kingston & Macmillan, 1995, Experiment I), shown that variation in F1 and nasality are not privileged relative to simultaneous variation in both (Kingston & Macmillan, 1995, Experiment II), and argued that integrality results at least partly from decisions (unconsciously) made by listeners (Macmillan et al., 1999). While the current study’s Experiment 2 was structured as a replication of Kingston and Macmillan’s (1995) Experiment I, the primary objective of Experiment 2 was different. Rather than demonstrating the existence of integrality within a population of native English listeners, Experiment 2 was intended to measure the degree of integrality within individual participants. More generally, the current

study is intended as an investigation of the connection between integrality in perception and covariation of phonetic properties in production rather than an investigation of integrality itself.

The results of the within-participant tests of the production-perception correlation were inconclusive. However, the individual results of the two experiments were inconsistent with the enhancement of contrasts at a language-wide or per-phoneme level but compatible with the enhancement of contrasts between individual lexical items.

The paper is structured as follows. §2 describes Experiment 1 and its results. §3 describes Experiment 2 and its results. §4 is the general discussion of results from both experiments and their implications for the enhancement hypothesis. §5 is the conclusion.

2 Experiment 1: Covariation of F1 and nasality in production

Experiment 1 assessed the degree of covariation between F1 and nasality that each participant displayed in their productions. Throughout this section, ‘vowel’ refers to a vowel in a primary stressed syllable unless otherwise noted.

2.1 Methods

2.1.1 Participants

The participants ($n = 19$) were native speakers of English comprising undergraduates recruited via the UC Santa Cruz Department of Linguistics’ online Linguistics Experiments Participant Sign-Up system and via word-of-mouth, graduate students recruited via word-of-mouth, and the experimenter. They were compensated \$10 for their participation. Two participants’ recordings had

technical issues which made their data unusable, meaning that for the purposes of data analysis $n = 17$.

2.1.2 Materials

The materials comprised a word list in which three factors were manipulated. Each factor had two levels, for a total of $2 \times 2 \times 2 = 8$ conditions. The factors are given below.

- i. Postvocalic consonant nasality: oral (VC), nasal (VN).
- ii. Vowel height: $V_{h(igher)}$, $V_{l(ower)}$, where V_h is higher than V_l , e.g., $[i, \varepsilon]$ or $[u, \Lambda]$. (Backness is held constant within pairs, i.e., V_h and V_l share the same backness, so $[i, \Lambda]$ would not form a usable V_h, V_l pair.)
- iii. Possible to construct minimal pairs by altering factors (i) and/or (ii): possible, not possible.

The word list was divided into 40 items of six words each (240 item words in total) and 240 individual filler words. The six-word items were divided into a ‘quartet’ of four words and two ‘loners’. The quartet and loner words differed in factor (iii), the possibility of constructing minimal pairs by altering the height of the vowel and/or the nasality of the postvocalic consonant: for the quartets, it was possible, and the quartets consisted of the four resulting words, e.g., *lift*, *left*, *lint*, *lent*; for the loners, it was not possible, e.g., *theft*, **thent*, **thift*, **thint*.

The quartet words varied in factor (i), the nasality of the consonant immediately following the vowel, and factor (ii), the height of the vowel. Consequently, each item contained a quartet with words containing the sequences V_hC , V_lC , V_hN , and V_lN , where V_h and V_l are vowels differing in height, C is an oral consonant, and N is a nasal consonant, crossing factors (i) and (ii). The loners crossed factors (i) and (ii) across items; each pair of loners contained two out of

V_hC , V_lC , V_hN , and V_lN , with equal numbers of each type for each pair of V_h and V_l .

	Quartet		Loners	
	V_h	V_l	V_h	V_l
VC	<i>lip</i>	lap	vision	factor
VN	limp	lamp	–	–
VC	lift	<i>left</i>	–	theft
VN	lint	lent	limbo	–
VC	hut	hot	such	–
VN	<i>hunt</i>	haunt	–	nonsense
VC	bead	bid	–	–
VN	bean	<i>bin</i>	cuisine	symptom
VC	hick	heck	<i>glitch</i>	skeptic
VN	hymn	hem	–	–
VC	newt	nut	–	<i>dozen</i>
VN	noon	nun	platoon	–

Table 1: Examples of items used in the production word list. Words in *italics* are an example of six words that could appear in a presentation list together.

To facilitate measurement of F1, the only vowels permitted were the monophthongs [i, ɪ, ε, æ, ɑ, ʌ, u] ([ʊ] is excluded because it does not occur before nasals and thus quartets involving it cannot be constructed; [ɔ] and [a] were treated as equivalent in the construction of items because the majority-Californian participants were assumed to have the cot-caught merger).

An attempt was made to construct at least eight quartets per pair of vowels. It was possible to find eight quartets for five pairs of vowels: [i, ɪ], [ɪ, ε], [ɪ, æ], [u, ʌ], and [ʌ, ɑ], for a total of 40 items containing 160 quartet words and 80 loners, 240 words total. The full list of items is given in Appendix A. The full list of 480 words was divided into six presentation lists of 80 words via Latin square such that no presentation list contained two words from the same item, nor did any list contain a pair of homophones. For example, given the items in Table 1, a single presentation list might contain the words *lip*, *left*, *hunt*, *bin*, *glitch*, *dozen*, or any other combination of six words from six different items

which differed from each other in at least one factor.

It has been shown that the magnitude of nasal coarticulation in English words is positively correlated with phonological neighborhood density (ND; Scarborough & Zellou, 2013). Consequently, in addition to the phonotactic and minimal pair conditions, the lexical factors of frequency and phonological neighborhood density were recorded for all words in the word list. These values were determined with reference to the Irvine phonotactic online dictionary (IPhoD; Vaden, Halpin, & Hickok, 2009). Because of the rarity of quartets of words like those used in this experiment, it was impractical to control frequency and ND for the quartet words. However, these factors *were* controlled for the loners and fillers. The loners and fillers were chosen such that their frequency distribution was similar to that of the quartet words. Because the presence and absence of minimal pairs is (for obvious reasons) correlated with neighborhood density, the loners systematically had lower ND than the quartet words. The fillers were selected such that their ND distribution resembled that of both the loners and quartet words as far as was possible, though this still produced an ND distribution resembling that of the loners more than that of the quartet words. Figure 1 shows the distributions of frequency and neighborhood density for each type of word.

Rough matching of neighborhood density and frequency was accomplished as follows. The quartet words were divided into eight $\log_{10}$ frequency ($\log_{10}WF$) bins of equal width, and the number of words n in each bin counted. Candidate loners were divided into the same bins, and $n/2$ loners were drawn from each bin. If n for a certain bin was an odd number (causing $n/2$ to be a fraction), then 1 was subtracted from that bin's n and added to the n of the nearest bin with a smaller, odd n . For example, given two bins with $n_1 = 17$ and $n_2 = 9$, these values would be adjusted to $n_1 = 16$ and $n_2 = 10$. For these two bins, 8

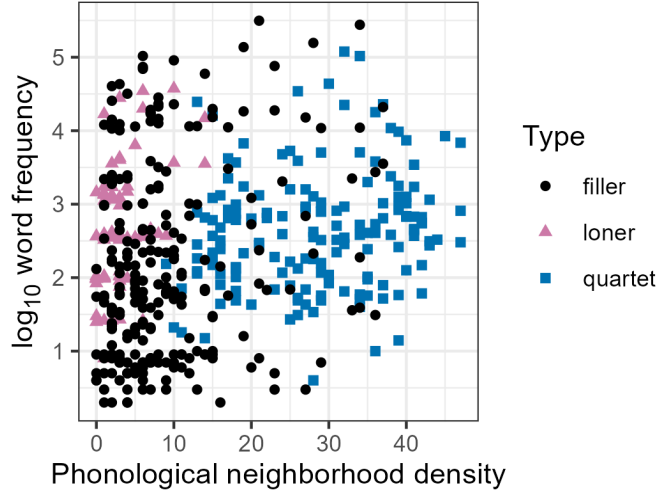


Figure 1: Distribution of $\log_{10}$ word frequency based on the SUBTLEXus database (Brysbaert & New, 2009) and phonological neighborhood density based on IPhoD (Vaden, Halpin, & Hickok, 2009) for filler, loner, and quartet words.

and 5 loners would be drawn, respectively.

To generate fillers, the IPhoD database was divided into four frequency bins ($\log_{10}\text{WF} < 1$, $1 \leq \log_{10}\text{WF} < 2$, $2 \leq \log_{10}\text{WF} < 4$, and $4 \leq \log_{10}\text{WF}$) and three neighborhood density bins ($\text{ND} \leq 4$, $5 \leq \text{ND} \leq 9$, and $10 \leq \text{ND}$), and each intersection of frequency and neighborhood density bins populated with an equal number of monosyllabic and disyllabic words. If there were not enough words in the IPhoD database that matched the criteria for one of these intersections, the remaining slots in that intersection were divided as close to equally as possible among the other intersections and populated with words that met their criteria.

2.1.3 Procedures

Before the experiment, participants completed a consent form. After the experiment, they completed a debriefing form asking them about their experience in the experiment and any thoughts they had about it. At this point, each partic-

ipant was offered the opportunity to also participate in Experiment 2. If they declined, they immediately completed a demographic questionnaire and read a short document explaining the purpose of the experiments and the hypotheses being tested. Otherwise, this final questionnaire and document were presented to them after they had completed Experiment 2.

To enable within-participant analysis of correlations between production and perception data, each participant was assigned a unique six-digit numeric ID code and the data they produced in Experiment 1 and Experiment 2 if they participated was associated with this code. Participants' names were associated with the code for the duration of their participation; after their last session, their names were deleted, anonymizing their data.

Recording was done in a quiet room on the UC Santa Cruz campus. Oral and nasal air pressure were recorded using a Glottal Enterprises (GE) oronasal airflow mask in which the mouth and nose are placed in separate chambers, each of which is equipped with GE PT-2E pressure transducer. Participants held the mask by an attached handle and raised it to their faces as necessary. Disregarding some vibrations transmitted through the silicone mask itself, this allows for mostly separate recordings of audio signals emitted from the mouth and nose. The relative average amplitudes of these recordings for a given time window can then be compared to determine the relative nasality in that time window (see §2.1.4 for details).

The pressure transducers were connected to a GE MS-110 pressure transducer unit which was in turn connected via USB adapter to a computer. The output of the MS-110 was recorded at a sampling rate of 11,025 Hz in Audacity (Team, 2025); this sampling rate was necessary to avoid picking up an 8 kHz hum and accompanying harmonics which are present in the MS-110's output. Simultaneously, audio recordings were made using an Audio-Technica

AT831b cardioid condenser lavalier microphone connected to a Zoom H5 solid-state portable recorder at a 44,100 Hz sampling rate.

Participants were assigned presentation lists in the order in which they participated. They were instructed to speak normally, as if talking to a friend, and to repeat items if they believed they had spoken disfluently. At the beginning of the recording, they read their six-digit participant ID code aloud to facilitate later identification of the recordings. The 80 words from the presentation list were presented one at a time on a computer screen in randomized order; participants read each word aloud and pressed a key to advance to the next until all 80 words had been read, at which point there was a break. After this break, the 80 words were presented a second time in re-randomized order, for a total of 160 tokens per recording. The recording sessions lasted approximately 5 minutes on average.

2.1.4 Measurements

The lavalier and oronasal recordings were synchronized in Audacity by aligning landmarks in the recordings. To facilitate this process, participants were instructed to clap at the beginning, midpoint, and end of the recording session, producing three impulses which were clearly visible in the recordings' waveforms. For each of the three impulses, the offsets between the initial zero-crossing of the largest positive-going peaks in the lavalier recording and the oral channel of the oronasal recording were measured, and the mean and standard deviation of these offsets calculated (mean SD = 1 ms, max SD = 5 ms). The mean offset was then subtracted from one of the recordings by excising a portion from the beginning, and the alignment double-checked by listening for audible distortion in words near the beginning, midpoint, and end of each set of recordings.

To facilitate automatic text grid annotation of the recordings, all parts of the recordings other than the participant reading the word list were excised,

including silences corresponding to the beginning, break, and end of the session, extraneous speech, and the participant’s ID code. If a participant repeated a word, the last repetition was kept and the prior ones deleted.

Also deleted were tokens involving obvious mispronunciations or tokens for which there was distortion in one or more channels. All cuts were made at zero crossing points of the waveform. In total, 80 tokens out of 2,720 or 2.94% were cut, leaving 2,640 tokens produced by 17 speakers. 71 of the cut tokens were from three participants (550504, 571700, and 904221) and were cut due to noise. The remaining 9 were cut due to mispronunciation (specifically, [ˈæbət] for *abet*, [ˈæsəs] for *assess*, [bɹusˈkeɪ] for *brusque*, [hæm] for *hymn*, [stɹɪn] for *strewn*, and [tɹɒʊ] for *trough*.)

The intensity of the lavalier and oronasal recordings was normalized using Audacity’s **Normalize** function such that the maximum intensity of each recording was -1 dB. The two channels of the oronasal recording were not normalized independently, i.e., the intensity of both channels was adjusted the same amount, preserving their relative intensities.

Initial annotation was performed with the Montreal Forced Aligner (MFA; McAuliffe et al., 2017) equipped with the English (US) ARPA dictionary v3.0.0 (McAuliffe & Sonderegger, 2024) and taking the lavalier audio and a text transcription as input. The resulting segment intervals were manually adjusted in Praat (Boersma & Weenink, 2023). Words that the forced aligner failed to segment were entirely manually annotated; the transcriptions used for each word were taken from IPhOD. The boundaries of vowels were identified by the appearance or disappearance of formant structure if the adjacent consonant(s) lacked formants, or by large changes in formant structure if the adjacent consonant(s) had formants.

Unless otherwise noted, data analysis was conducted in R (R Core Team,

2023). To measure nasality, separate intensity contours were extracted from the oral and nasal channels of the oronasal recordings using a Praat script, read into R, and normalized to a range of [0, 1] for each speaker (for details, see Bennett et al., 2025). Each vowel was divided into 17 time windows of equal duration, and the nasalance or % nasal of each window was calculated using the formula $\text{Nasalance} = 100(A_N / (A_N + A_O))$, where A_N = the average amplitude of the nasal channel for a given time window and A_O = the average amplitude of the oral channel for the same time window. This yielded 44,880 observations, of which 83 (0.18%) were dropped due to a lack of airflow data precluding the calculation of nasalance, leaving 44,797 total nasalance observations.

To measure F1, a Praat script (Styler, 2020) was used to observe the formants of each vowel at 17 evenly spaced 5 ms windows. To facilitate comparison of F1 values with nasalance and across speakers, F1 values were normalized to a range of [0, 1] for each speaker using the `scales` package (Wickham, Pedersen, & Seidel, 2023). F1 observations corresponding to the time windows of the dropped nasalance observations were also excluded, leaving 44,797 F1 observations. Due to the nature of the statistic used to estimate the correlation between F1 and nasalance, it was important that there be the same number of F1 and nasalance observations, or rather that each time point for each vowel had both corresponding F1 and nasalance observations. References to ‘F1’ in the subsequent analysis should be understood as referring to normalized F1 unless otherwise stated.

2.2 Results

2.2.1 Data characteristics and segmental processes

Raw F1 values varied between 58 and 1,423 Hz, with a mean value of 603 Hz. Within speakers, F1 displayed bi- or trimodal distributions in oral contexts,

corresponding to different vowel heights. The number of modes that F1 distributions displayed in nasal contexts was more variable (Figure 2).

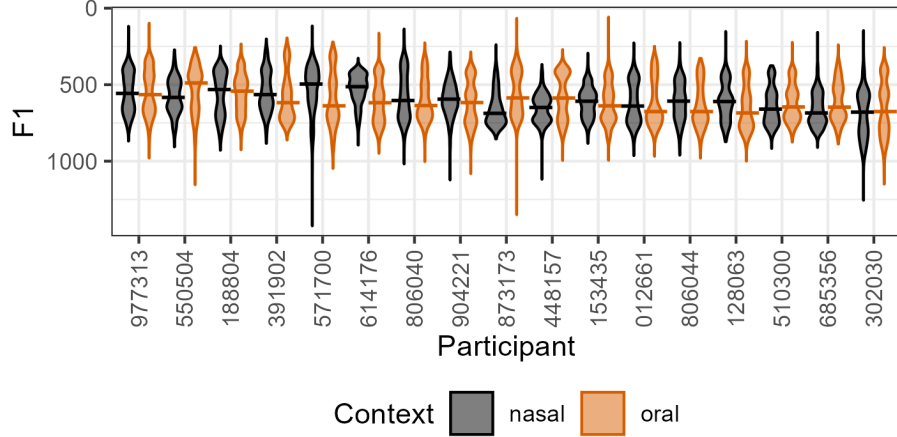


Figure 2: Violin plot of raw F1 values in vowels before nasal consonants (black) and oral consonants (orange) for each participant. Horizontal bars indicate median F1 values. The corresponding normalized F1 distributions have the same shape, but ‘stretched’ or ‘compressed’ to fit a scale of 0 to 1.

Nasalance values varied between .103 and .695, with a mean value of .437. Within speakers, the distribution of nasalance was generally bimodal, with the difference between modes corresponding to whether a vowel appeared before a nasal or oral consonant (Figure 3).

The data were checked for evidence of three processes active in California English which could affect F1 values: the raising of /æ/ before nasal consonants, particularly /ŋ/ (Brotherton et al., 2019; Zellou, Scarborough, & Kemp, 2020), and the BIT/BET and COT/CAUGHT mergers (Cardoso et al., 2017, and citations therein).

A visual inspection of Figure 4 shows that for the speakers in this study, /ɑ/ and /ɔ/ are merged. Consequently, in subsequent analysis these two vowels are treated as a single vowel /ɑ/. Meanwhile, /ɪ/ and /ɛ/ are not merged, though their F1 ranges overlap with each other and with that of /ɑ/. These vowels are

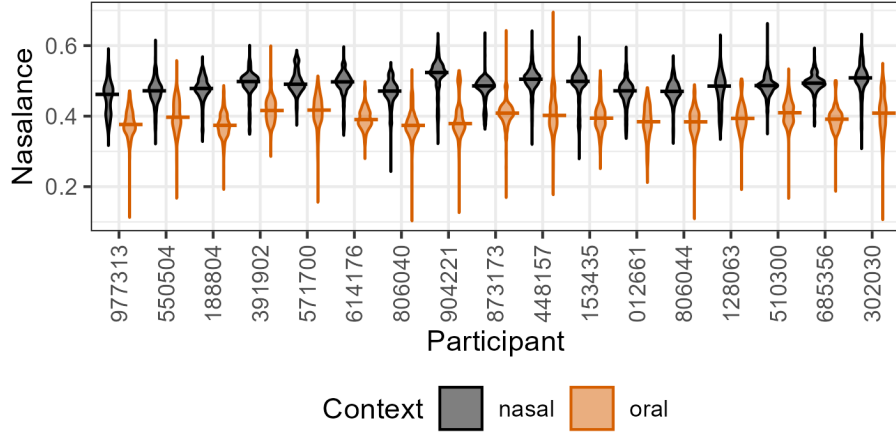


Figure 3: Violin plot of normalized nasalance values in vowels before nasal consonants (black) and oral consonants (orange) for each participant. Horizontal bars indicate median nasalance values.

treated separately in subsequent analysis.

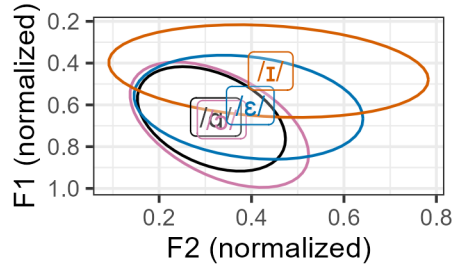


Figure 4: Mean realizations for /ɪ/ (orange), /ε/ (blue), /ɔ/ (pink), and /ɑ/ (black) across speakers. Ellipses represent a 95% confidence interval assuming that individual points are part of a multivariate t -distribution.

A visual inspection of Figure 5 shows that /æ/ is raised before nasal consonants and especially before /ŋ/. The variation in /æ/’s normalized F1 appears comparable to that of /u/ or /ε/, with the exception of the sequence /æŋ/. It is possible that /æŋ/ sequences might act as a source of negative covariation between F1 and nasalance, thus introducing a bias in the final estimation of the F1-nasalance correlation. However, there are two arguments against excluding

/æŋ/ sequences.

First, observations of /æŋ/ sequences comprise just 0.91% of the data, meaning that any bias they introduced would likely be negligible. Second, there is to the best of my knowledge no principled reason to believe that native English listeners afford special perceptual treatment to /æŋ/ sequences relative to other VN sequences, and the omission of a specific /VN/ sequence would thus reduce the ecological validity of the analysis. Consequently, observations of /æŋ/ sequences were retained.

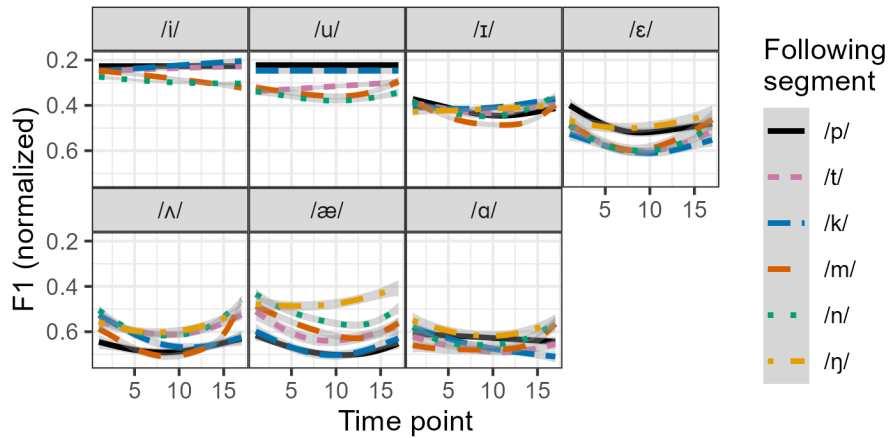


Figure 5: Mean normalized F1 trajectories for each vowel before /p, t, k/ and /m, n, ŋ/. Lines were fitted using locally estimated scatterplot smoothing (LOESS). Shaded areas represent a 95% confidence interval.

The data were also checked for differences in nasalance conditioned by the syllable role of the following consonant, i.e., whether it was tautosyllabic with the vowel. A visual inspection of Figure 6 suggests that while the syllable role of the following consonant did have some effect on nasalance in the vowel, these effects were smaller for nasal consonants than for oral consonants. Consequently, the syllable role of the following consonant was not considered further in the analysis of nasalance.

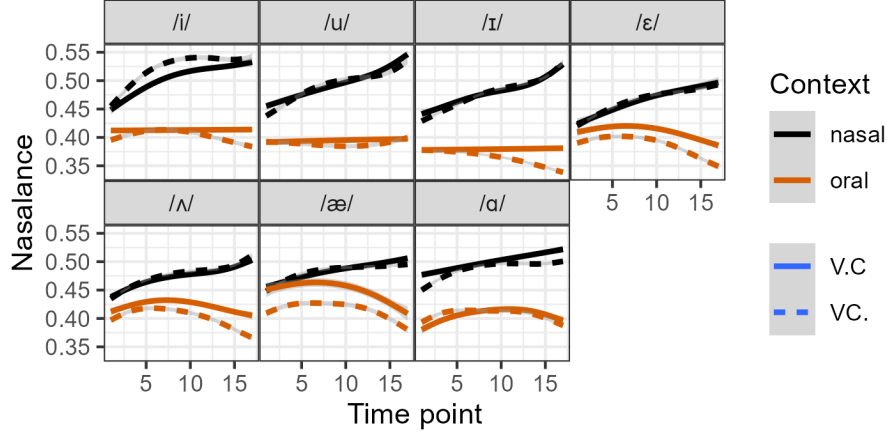


Figure 6: Mean nasalance trajectories for each vowel before tautosyllabic (‘VC.’) and heterosyllabic (‘V.C’) oral and nasal consonants. Lines were fitted using LOESS. Shaded areas represent a 95% confidence interval.

2.2.2 F1-nasalance correlation: effects of vowel quality and context

Spearman’s rank order correlation test ρ_s was used to estimate the correlation of nasalance and normalized F1. In this test, the data are first transformed from absolute values to ascending ordinal ranks. For example, the set of values [.13, .40, .66, .91] would be transformed to [1, 2, 3, 4]. Pearson’s correlation test is then performed on the rankings, producing a value ranging from 1 (perfect covariance of rankings) to -1 (perfect negative covariance), with 0 representing no covariance. Spearman’s test was used because the data are not bivariate normally distributed, violating a critical assumption of Pearson’s test.

Additionally, the kurtosis of the normalized F1 data (min: -1.30 , median: $.73$, max: 1.49) and nasalance data (min: $-.50$, median: 1.96 , max: 6.58) are sufficient to cast doubt on the reliability of confidence intervals (CIs) derived from Pearson’s test. CIs were therefore calculated using Fisher’s z' transformation as modified by Fieller, Hartley, and Pearson (1957), as the combination of this method with Spearman’s test produces CIs which are relatively robust

against nonnormality (Bishara & Hittner, 2017, p. 302). All CIs were calculated with a confidence level of 95% unless otherwise stated. Full tables of correlations are given in Appendix B.

The effects of vowel quality and the (non-)nasality of the postvocalic consonant on normalized F1 and nasalance are shown in Figure 7. A positive correlation between normalized F1 and nasalance was found for most vowels in oral contexts; the smallest statistically significant correlation was for /æ/, $\rho_s(2, 373) = .05$, $p < .05$, CI [.01, .09], and the largest was for /ʌ/, $\rho_s(3, 865) = .19$, $p < .0001$, CI [.16, .22]. The only vowel to show a negative correlation was /i/, $\rho_s(2, 261) = -.19$, $p < .0001$, CI [-.23, -.15]. Among the vowels in nasal contexts, four displayed statistically significant correlations of which three were positive (/u/, /ε/, and /ʌ/) and one negative (/i/).

A more detailed breakdown of correlations can be seen in Figure 8, which shows counts of correlations for each vowel in each context as produced by each participant. For many participants, non-high vowels in oral contexts tended to show positive F1-nasalance correlations, while correlations among high vowels in oral contexts were more varied but trended negative, especially for /i/. In nasal contexts, fewer participants' vowels achieved statistically significant F1-nasalance correlations, but those that did tended to show slightly more positive correlations than negative ones.

The variability in F1-nasalance correlations among vowels in nasal contexts may be related to the two different patterns of F1 context (in-)sensitivity among non-high vowels seen in Figure 9. For all speakers, high vowels tended to be (slightly) lowered in nasal contexts as compared to oral contexts. For some speakers such as 188804 this lowering effect tapers off either in the mid vowels or low vowels, meaning that for non-high vowels F1 values are relatively insensitive to nasality. However, for other speakers such as 391902 the lowering effect

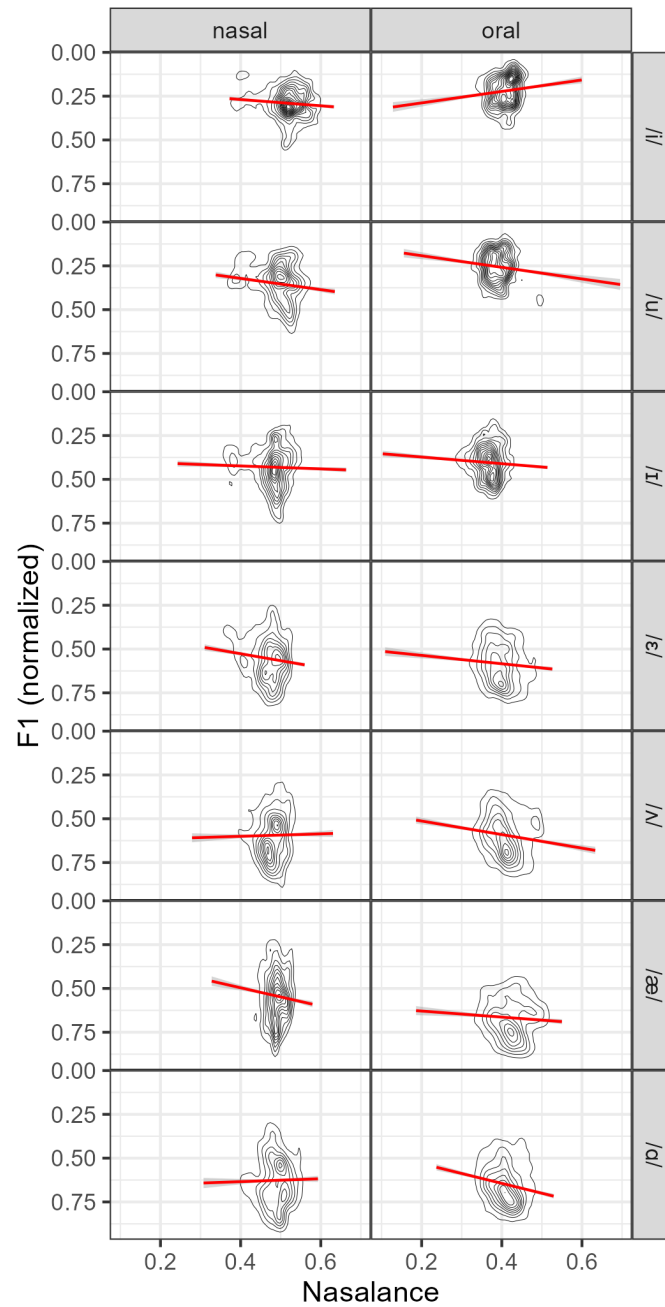


Figure 7: Density plot of normalized F1 values and nasalance for each vowel quality before oral or nasal consonants, plus best-fit correlation lines. Shaded areas represent a 95% confidence interval.

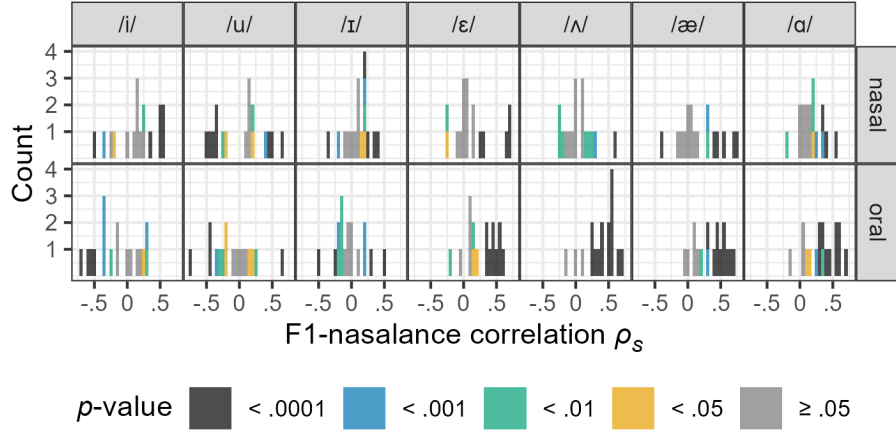


Figure 8: Histogram of F1-nasalance ρ_s correlation values for each vowel produced by each participant in oral and nasal contexts, colored by p -value of the correlation.

reverses among the low vowels, meaning that for all vowel qualities in nasal contexts F1 values trend towards the middle.

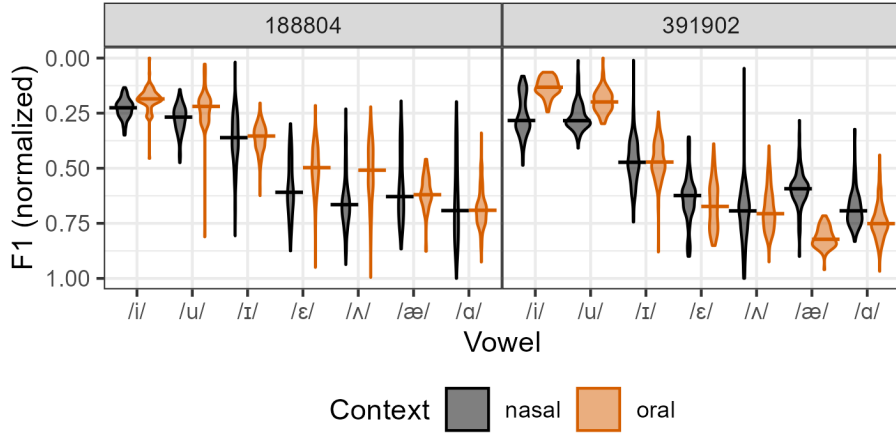


Figure 9: Violin plot of normalized F1 values for each vowel before nasal consonants (black) and oral consonants (orange) for selected speakers. Horizontal bars indicate median F1 values.

2.2.3 F1-nasalance correlation: effects of word type

‘Word type’ refers here to a word’s status as a filler, loner, or quartet member. The effects of word type are summarized in Figure 10, which shows F1-nasalance ρ_s values for each participant across types, and Figure 11, which shows the distributions of these values as a histogram. Across all observations, normalized F1 and nasalance displayed a very small but statistically significant positive correlation, $\rho_s(44,797) = .05$, $p < .0001$, CI [.04, .06]. Across all types of words and within participants, correlation estimates varied from large and negative ($\rho_s(2,717) = -.27$, $p < .0001$, CI [−.30, −.23]) to very large and positive ($\rho_s(2,720) = .46$, $p < .0001$, CI [.43, .49]), with a median value of $\rho_s(2,717) = .09$, $p < .0001$, CI [.06, .13].

The F1-nasalance correlations found among the fillers resembled those found for the whole data set, which is unsurprising given that the fillers made up half that data set. Across all participants, the F1-nasalance correlation was very small, $\rho_s(22,340) = .03$, $p < .0001$, CI [.01, .04], while individual participants’ correlations ranged widely (min: $\rho_s(1,360) = -.30$, $p < .0001$, CI [−.35, −.25]; median: $\rho_s(1,326) = .07$, $p < .01$, CI [.02, .18]; max: $\rho_s(1,360) = .50$, $p < .0001$, CI [.46, .54]).

F1-nasalance correlations within the loner words were more varied. Across all participants, no statistically significant correlation was found, $\rho_s(7,463) = .01$, $p \geq .05$, CI [−.01, .03]. However, 14 out of 17 participants showed statistically significant F1-nasalance correlations, and these ranged very widely (min: $\rho_s(439) = -.38$, $p < .0001$, CI [−.46, −.30]; median: $\rho_s(467) = .10$, $p < .05$, CI [−.19, .00]; max: $\rho_s(442) = .64$, $p < .0001$, CI [.58, .69]). The wideness of the range of correlations may be due to small sample sizes (median $N = 439$) compared to the fillers (median $N = 1,357$).

Correlations within the quartet members resembled those within the filler

words more closely. The F1-nasalance correlation across all participants was small, $\rho_s(14,994) = .09$, $p < .0001$, CI [.07, .10], though greater than that in the fillers. The correlations ranged less widely than in the loners and fillers (min: $\rho_s(918) = -.18$, $p < .0001$, CI [-.24, -.11]; median: $\rho_s(714) = .10$, $p < .01$, CI [.02, .17]; max: $\rho_s(918) = .38$, $p < .0001$, CI [.28, .40]).

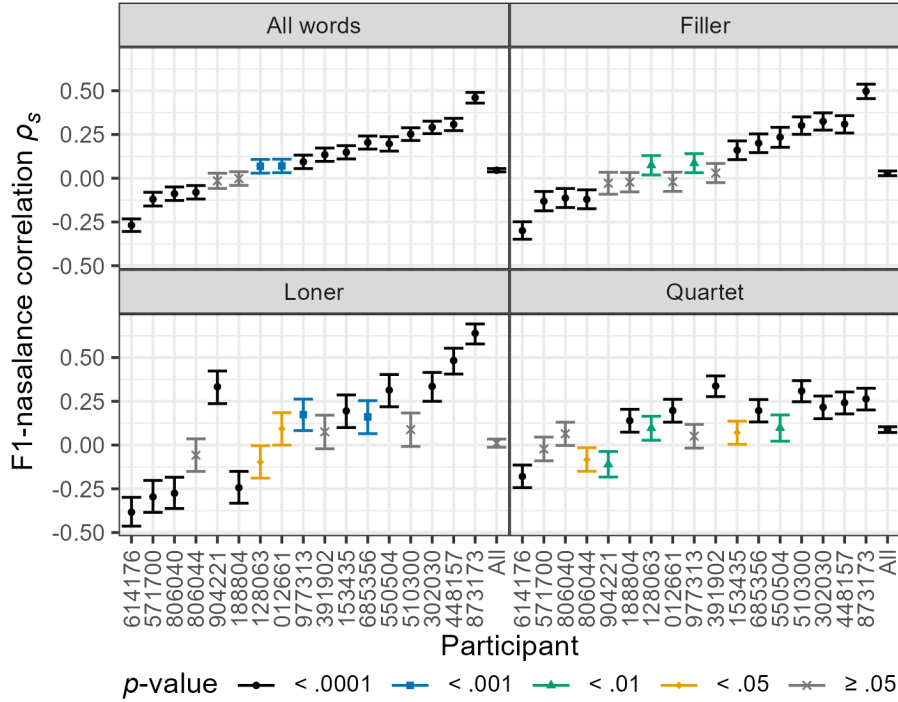


Figure 10: Normalized F1-nasalance correlation estimates ρ_s for each participant and for each type of word. Error bars represent a 95% confidence interval.

Overall, most speakers displayed positive F1-nasalance correlations across all word types, i.e., lower vowels were more nasalized and more nasalized vowels were lower. However, there was a minority of speakers for whom this trend reversed, i.e., lower vowels were *less* nasalized, and more nasalized vowels were higher. As was previously noted, some speakers pull vowels in nasal contexts towards the middle of the F1 range. Are the negative F1-nasalance correlations

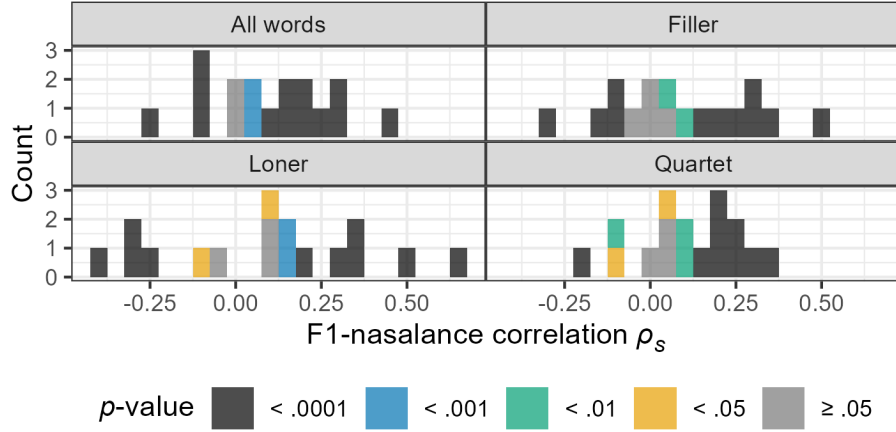


Figure 11: Histogram of F1-nasalance ρ_s values for each type of word produced by each participant, colored by p -value.

observed in this section attributable to these same speakers? To test this, the strength of the midward pull was estimated for each word type for each speaker by calculating the difference between mean normalized F1 values for /ɑ/ in oral and nasal contexts. The correlation between these estimates and the statistically significant ($p < .05$) F1-nasalance ρ_s values for the same speakers and word types was very large, $\rho_s(30) = .54$, $p < .01$, CI [.21, .76], indicating that many of the negative F1-nasalance correlations were contributed by speakers with the midward pull.

2.2.4 F1-nasalance correlation: effects of phonological neighborhood density and word frequency

Despite the finding that the range of F1-nasalance correlations varied across types of words, phonological neighborhood density did not have any appreciable effect on correlations between normalized F1 and nasalance, nor did word frequency. A visual inspection of Figure 12 reveals no apparent pattern in the distribution of ρ_s , $\log_{10}$ WF, and ND values. A Spearman test for a correlation

between statistically significant ($p < .05$) normalized F1-nasalance ρ_s values and $\log_{10}\text{WF}$, which was divided into 50 bins of width 0.1, did not produce significant results: $\rho_s(50) = .20$, $p \geq .05$, CI $[-.10, .47]$. The same was true for a Spearman test for correlation between significant normalized F1-nasalance ρ_s values and ND, $\rho_s(47) = .04$, $p \geq .05$, CI $[-.26, .33]$.

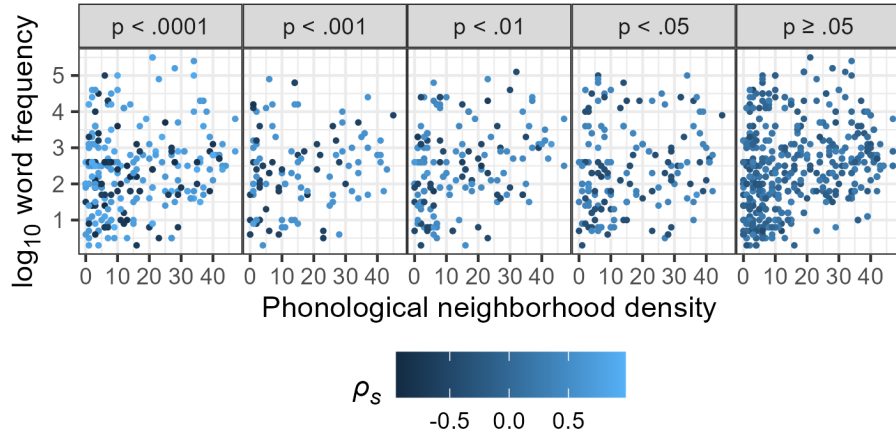


Figure 12: Scatter plots of Spearman correlations between normalized F1 and nasalance for every word in the data set compared to the word’s $\log_{10}$ frequency and phonological neighborhood density, faceted by the p -value of the correlation.

Linear mixed-effects models (LMEs) estimated with restricted maximum likelihood (REML) were used to further assess the relative magnitude of word frequency and ND’s effects on F1, nasalance, and the F1-nasalance correlation ρ_s . The strategy used to gauge the impact of each variable was to fit three LMEs, one each with normalized F1, nasalance, and F1-nasalance ρ_s as the dependent variable. The relative effect sizes of each of the fixed effects were then compared, including ND, $\log_{10}\text{WF}$, and their interaction (interactions between factors x and y are denoted $x \times y$).

All LMEs had participant as a random effect. This allows for generalizations to be made across participants without the risk of misidentifying an idiosyncrasy of the participants as a genuine effect. All calculations were done using the `lme4`

package (Bates et al., 2015).

A note on significance. While p -values are provided in Tables 2, 3, and 4, they should be regarded with some skepticism. The interaction between fixed and random effects in an LME results in a violation of the assumptions underlying traditional significance tests. Consequently, p -values derived from LMEs are at best inaccurate and at worst totally meaningless (Baayen, Davidson, & Bates, 2008, p. 396), and I do not discuss p -values when reporting the results. As an alternative, CIs and the t statistic (the ratio of effect size to its standard error) are included. Larger t values correspond to narrower CIs relative to effect size and decreasing likelihood that an observation is due to chance.

To estimate the relative magnitude of ND and $\log_{10}$ WF's impacts on F1, an LME was fitted with normalized F1 as the dependent variable, nasalance, $\log_{10}$ WF, ND, and the $\log_{10}$ WF $\times$ ND interaction as fixed effects, and participant as a random effect; results are shown in Table 2. The intercept, .404, is the predicted normalized F1 value when nasalance, $\log_{10}$ WF, and ND are all 0; it is smaller than the median normalized F1 value of the data set, .501, by about 10%. The positive coefficient of nasalance indicates that increases in nasalance result in increases in F1, i.e., they are positively correlated, and the large t -value (comparable in magnitude to the t -value of the intercept) indicates that this effect is likely to be significant, in accordance with the results of the Spearman tests previously reported. Compared to the coefficient of nasalance, the coefficients of $\log_{10}$ WF, ND, and their interaction are two orders of magnitude smaller. This suggests that if $\log_{10}$ WF, ND, or their interaction have any effect, it is extremely small compared to that of nasalance. In other words, any effects of frequency and neighborhood density on vowel height were negligible.

To estimate the relative impacts of $\log_{10}$ WF and ND on nasalance, an LME was fitted with nasalance as the dependent variable, normalized F1, $\log_{10}$ WF,

Dependent variable: F1 (normalized)				
Parameter	Estimate	CI	<i>t</i> -value	<i>p</i> -value
Intercept	.404	.382, .426	18.13	< .0001
Nasalance	.216	.203, .228	17.29	< .0001
$\log_{10}\text{WF}$	-.004	.003, .005	-4.20	< .0001
ND	-.003	-.003, -.003	-14.99	< .0001
$\log_{10}\text{WF} \times \text{ND}$	.001	.001, .001	18.06	< .0001

Table 2: Results of LME using formula **F1 (normalized) $\sim$ nasalance + log10WF * ND** with participant as a random effect; $N = 1,356$.

ND, and the $\log_{10}\text{WF} \times \text{ND}$ interaction as fixed effects, and participant as a random effect; the results are shown in Table 3. The intercept, .424, is the predicted nasalance when normalized F1, $\log_{10}\text{WF}$, and ND are all 0; it is close to the median nasalance value in the data set, .437. Normalized F1 has a positive coefficient and a moderately large *t*-value, indicating a positive correlation which is likely to be statistically significant. Meanwhile, the coefficients of $\log_{10}\text{WF}$, ND, and their interaction were zero or almost zero, suggesting that they had no effect on nasalance.

Dependent variable: nasalance				
Paramter	Estimate	CI	<i>t</i> -value	<i>p</i> -value
Intercept	.424	.420, .428	117.22	< .0001
F1 (normalized)	.031	.029, .032	17.21	< .0001
$\log_{10}\text{WF}$	.002	.001, .002	3.90	< .0001
ND	.000	.000, .000	-.52	> .05
$\log_{10}\text{WF} \times \text{ND}$	.000	.000, .000	-5.28	< .0001

Table 3: Results of LME using formula **nasalance $\sim$ F1 (normalized) + log10WF * ND** with participant as a random effect; $N = 1,356$.

To estimate the magnitude of the effects of $\log_{10}\text{WF}$, ND, and their interaction on the F1-nasalance correlation itself, mean normalized F1, mean nasalance, and Spearman ρ_s and *p* values were calculated for each word (averaging across tokens) produced by each speaker, yielding a mini-data set of 1,356 observations. An LME was fitted to this mini-data set with F1-nasalance ρ_s as the dependent variable, $\log_{10}\text{WF}$, ND, and their interaction as fixed effects, and participant as

a random effect; the results are shown in Table 4.

The intercept, .051, is the expected F1-nasalance ρ_s value when $\log_{10}\text{WF}$ and ND are both 0. The coefficients of $\log_{10}\text{WF}$, ND, and their interaction are very small and their CIs cross 0; their t -values are also very small. Altogether this indicates a high degree of uncertainty. If there was an effect, it was very hard to detect; alternatively, $\log_{10}\text{WF}$, ND, and their interaction truly had no effect on the F1-nasalance correlation.

Dependent variable: F1 (normalized)-nasalance ρ_s					
Parameter	Estimate	CI	t -value	p -value	
Intercept	.051	-.001, .103	.98	$\geq .05$	
$\log_{10}\text{WF}$	.012	-.004, .029	.12	$\geq .05$	
ND	.002	.000, .005	.91	$\geq .05$	
$\log_{10}\text{WF} \times \text{ND}$	.000	-.001, .001	.15	$\geq .05$	

Table 4: Results of LME using formula $\rho_s \sim \log_{10}\text{WF} * \text{ND}$ with participant as a random effect; $N = 715$.

2.3 Discussion

An experiment was conducted in which 17 native speakers of English read a word list while nasalance and F1 measurements were taken. Half the words were fillers, and the other half items. Words in items systematically varied in

- i) the height of the vowel in the primary stressed syllable, e.g., *messed*, *mist*,
- ii) the (non-)nasality of the postvocalic consonant, e.g., *messed*, *meant*, and
- iii) whether it was possible to construct minimal pairs by varying the first two factors, e.g., *messed*–*missed*–*meant*–*mint* vs. *festive*–**fistive*–**fentive*–**fintive*.

The filler words were chosen so that their frequency and phonological neighborhood density approximated that of the non-fillers.

Across all observations, there was a very small positive correlation between F1 and nasalance ($\rho_s = .05$), i.e., lower meant more nasalized and vice versa. A small effect size like this might seem to be of no practical importance, but small

effects may still have practical consequences if the effect cumulates over time (Funder & Ozer, 2019). For example, Abelson (1985) found that a professional baseball player’s batting skill (their batting average) accounted for just one third of 1% of the variance in their batting performance (the likelihood of getting a hit in a given at bat). (Coincidentally, this percentage of variance corresponds to a Pearson’s correlation of $r = .056$, not far off from the F1-nasalance correlation of $\rho_s = .05$.) This contrasts sharply with most people’s intuitions about the importance of batting skill to batting performance and game outcomes. Funder and Ozer give other examples of effects which seem obvious but have surprisingly small correlation values, e.g., the effect of antihistamines in relieving allergies ($r = .11$), the tendency of men to weigh more than women ($r = .26$), or the tendency of temperature to decrease with altitude ($r = -.34$).

The resolution to the apparent paradox is that an at bat is a single event, while observed batting skill and game outcomes are the aggregate of many such events, i.e., the effects of batting skill on at bats (and from there on game outcomes) are cumulative. Similarly, the effectiveness of an antihistamine pill, the weight of a person, or the temperature at a location are all single observations, and the ‘obvious’ trends we take for granted are the aggregate of many such observations. Over the course of a day, a listener might hear thousands or tens of thousands of vowels; over the course of weeks or months, millions. It would be surprising if this were not a large enough sample size for cumulative effects of even a slight F1-nasalance covariation to appear.

Even ignoring cumulativity, most speakers also displayed F1-nasalance correlations considerably larger than $\rho_s = .05$. Within speakers, 11 showed a positive F1-nasalance correlation ranging in strength from very small ($\rho_s = .07$) to extremely large ($\rho_s = .46$). Four other speakers showed negative F1-nasalance correlations varying less widely in size from very small ($\rho_s = -.08$) to large

($\rho_s = -.27$); for these speakers, lower meant less nasalized and vice versa. Assuming that listeners track individual F1-nasalance correlations for each of their (regular) interlocutors, it would be very surprising if these trends went unnoticed.

There was also considerable variation in the size of the F1-nasalance correlation across vowel qualities. The correlation was positive and largest in the high vowels /i/, /u/, and tended to diminish among /ɪ/, /ɛ/, /ʌ/ before disappearing among the low vowels /æ/, /ɑ/. In other words, in nasal contexts the high vowels tended to lower, the low vowels stayed the same, and the remaining vowels were a toss-up. However, a minority of speakers showed a negative F1-nasalance correlation for the low vowels /æ/, /ɑ/, i.e., rather than the lowering of vowels in nasal contexts simply tapering off among the low vowels, it instead reversed, with low vowels in nasal contexts being *higher* than in oral contexts. There was a very large correlation between the tendency to show this ‘midward pull’ in the low vowels and negative F1-nasalance correlations ($\rho_s = .54$), suggesting that for the four speakers showing a negative F1-nasalance correlation, the midward pull in the nasalized low vowels is responsible.

The size of F1-nasalance correlations also varied considerably by word type. Whether positive or negative, they tended to be more extreme among loners compared to quartet members and filler words. In other words, whatever effect nasalization had on height or vice versa in a given word, it was smaller when that word was part of a minimal pair which depended on vowel height, e.g., *pit-pet*, or the nasality of the postvocalic consonant, e.g., *pit-pin*, and greater when the word was not part of any such minimal pair, e.g., *distance*–**destance*–**dintance*.

The fact that loners had many fewer neighbors than quartet members on average, 2.95 vs. 28.30 respectively (see also Figure 1), might lead one to suspect that increasing F1-nasalance covariation is conditioned by decreasing neighbor-

hood density. However, a set of LMEs indicated that a word’s neighborhood density had no systematic relationship with the height or nasalization of the vowel in its primary stressed syllable nor with the correlation between those properties across words, i.e., ND had no apparent effects. (These LMEs also showed that frequency had no effect. While the realization of a word may be affected by both its frequency (Gahl, 2008) and ND (Scarborough & Zellou, 2013), the following discussion focuses on ND as that was the factor most closely related to word types as manipulated in Experiment 1.)

This result is surprising from the perspective of Hypo- and Hyperarticulation (H&H) Theory (Lindblom, 1990), which posits that the articulatory effort expended on a word is proportional to that word’s unpredictability in the current speech context. More precisely, the H&H model posits that the less information (in the information theoretic sense) about a word’s identity can be derived from context, the more information must be supplied by the speaker, resulting in the speaker hypo- or hyper-articulating proportionally to the richness of information sources other than the speech signal itself. This information theoretic view implies that any covariation between two phonetic dimensions (including F1 and nasality) in excess of what is physiologically necessary amounts to (theoretical, perhaps not practical) information loss. Consequently, increasing covariation relative to some standard is a form of hypoarticulation, and decreasing covariation a form of hyperarticulation.

The more neighbors a word has, the more competition it faces during word recognition (Vitevitch & Luce, 1999). Increasing the intensity of vowel nasalization can be understood as a way to exaggerate the difference between VC and VN sequences which distinguish neighboring words and thus as a form of hyperarticulation. Evidence supporting precisely this kind of hyperarticulation was found by Scarborough and Zellou (2013), who examined anticipatory nasal

coarticulation in speakers producing CVN(C) and CVC words from two ND bins (high- and low-ND) in five different speech styles. Using an acoustic measure of nasality, A1 – P0 (Chen, 1997), they found that nasalization was greater for speech styles requiring greater ‘clarity’ (e.g., speaking to an imagined hard-of-hearing listener) and for high-ND words.

Given this evidence that ND affects the magnitude of anticipatory nasal coarticulation in English and the much greater average ND of quartet words as compared to loners, why did loners have more extreme F1-nasalance correlations (both positive and negative) compared to quartet members when ND itself had no effect on F1, nasalance, or their correlation? In other words, why does one kind of difference (loner vs. quartet word) condition hypo- or hyperarticulation while a related difference, ND, does not?

Assuming that the difference between loners and quartet items is not spurious, it must be the case that hypo- and hyperarticulation of the F1-nasalance covariation are conditioned not by the number of neighbors, but the kind. By definition, the quartet words have neighbors which differ from them in vowel height and the nasality of the postvocalic consonant, while the loners do not. This means that the amount of F1 and nasalance information necessary to distinguish a word from its neighbors is greater for quartet words than for loners. The consequent prediction under H&H theory is that quartet words are less tolerant than loners of the information loss created by F1-nasalance covariation, whether positive or negative, prompting speakers to hyperarticulate and separate F1 and nasality to a greater degree in quartet words than in loners.

This interpretation is compatible with the results obtained by Scarborough and Zellou (2013). While they attributed differences in the magnitude of vowel nasalization to neighborhood density, the stimulus words were all monosyllables and ND was treated as a factor with just two levels (high and low), meaning that

many of the high-ND stimuli had neighbors which differed only in the nasality of the postvocalic consonant. In other words, each word's total number of neighbors was conflated with the presence or absence of neighbors which differed from it based on nasality. In light of the results of Experiment 1, it seems likely that the differential nasalization found by Scarborough and Zellou should be attributed to the presence of neighbors differentiated by nasality rather than overall neighborhood density.

The existence of variation between speakers and words in the degree to which F1 and nasality are collapsed together in production appears incompatible with any version of the enhancement hypothesis which does not permit the covariation to be sensitive to word-level factors such as the presence of certain types of neighbors, e.g., set degrees of covariation for individual phonemes. However, production data alone is not sufficient to rule on a hypothesis which concerns a relationship between both production and perception. A fuller test of this hypothesis requires the comparison of the covariation of F1 and nasality in production with estimates of integrality in perception.

3 Experiment 2: Integrality of F1 and nasality in perception

This experiment was structured as a partial replication of Kingston and Macmillan's (1995) Experiment I. The method employed to measure the degree of integrality between F1 and nasality for each participant was the Garner paradigm (Garner, 1974). In this design, participants are presented with a stimulus array in which two physically separable dimensions are varied orthogonally to each other. The participants are asked to categorize these stimuli in a series of two-alternative forced choice (2AFC) tasks. Integrality is diagnosed when the

categorization of stimuli by their value on one dimension is systematically affected by their value on the other dimension or vice versa. A detailed description of the Garner paradigm is given in §3.1.2.

3.1 Methods

3.1.1 Participants

The participants in Experiment 2 ($n = 11$) were a subset of those who participated in Experiment 1, including the experimenter. The two participants excluded from Experiment 1 due to technical issues with their recordings were not among those who participated in Experiment 2. One participant did not proceed beyond the training phase of the experiment and is not included in the count of participants. Participants were compensated \$16.50 per hour (California minimum wage as of 2025) plus a bonus of \$30 for completing both experiments.

3.1.2 The Garner paradigm and materials

The Garner paradigm (Garner, 1974) is designed to assess the degree to which two physically orthogonal dimensions are perceptually integrated. To accomplish this, a 2×2 array of stimuli varying in these dimensions is assembled, and participants are asked to categorize various subsets of this array in a series of 2AFC tasks. These subsets may differ in just one dimension or in both simultaneously, and the 2AFC tasks may involve exposure to a pair of stimuli or to all four. Integrality is diagnosed if the way in which the dimensions are combined (or not) in the different subsets of stimuli used across tasks systematically affects performance, e.g., if positive covariation of F1 and nasality results in better performance than varying just F1 or just nasality.

The two dimensions whose integrality was assessed in Experiment 2 were

the frequency of F1 in Hz, and the separation between the first nasal pole and first nasal zero in Hz $N_0 - N_1$, subsequently referred to simply as N. The stimuli consisted of a 2×2 array of synthetic vowels similar to those employed by Kingston and Macmillan (1995) and Macmillan et al. (1999), and varied by roughly one just-noticeable difference (JND) on each dimension. A 1 JND difference was used in order to induce errors and avoid ceiling effects, facilitating the detection theoretic analysis discussed in 3.2.2.

The vowels were placed in a [b_n] frame, resulting in four stimuli of the shape [bVn]. The stimuli were 360 ms in duration, comprising a 140 ms vowel steady state, two 80 ms flanking consonants, and two 30 ms transition periods between the consonants and vowel steady state. They were generated using the Klatt synthesizer (Klatt, 1980) implemented in Praat based on the parameters given in Kingston and Macmillan (1995) (see Appendix C for the full Klatt synthesizer parameters) and their intensity was normalized to 70 dB using Praat’s **Scale intensity** function. Impressionistically, the vowels of the stimuli could be transcribed as [u, o, ũ, õ], and for convenience they will be referred to using these symbols from this point on. Seven phonetically trained listeners including the author reported impressionistically that it was possible but very difficult to discriminate between the stimuli. Spectrograms of the stimuli are given in Figure 13.

These stimuli were presented in eight conditions or ‘tasks’ as they are traditionally called in the Garner paradigm. Three types of tasks were used: baseline, correlated, and selective attention. These tasks differed in the subsets of stimuli used in each. The four baseline tasks involved the categorization of two stimuli which differed either in F1 or N but not both, e.g., [u] vs. [o] or [u] vs. [ũ]. The two correlated tasks involved the categorization of two stimuli which differed in both F1 and N simultaneously: [u] vs. [õ] and [ũ] vs. [o]. The former pair

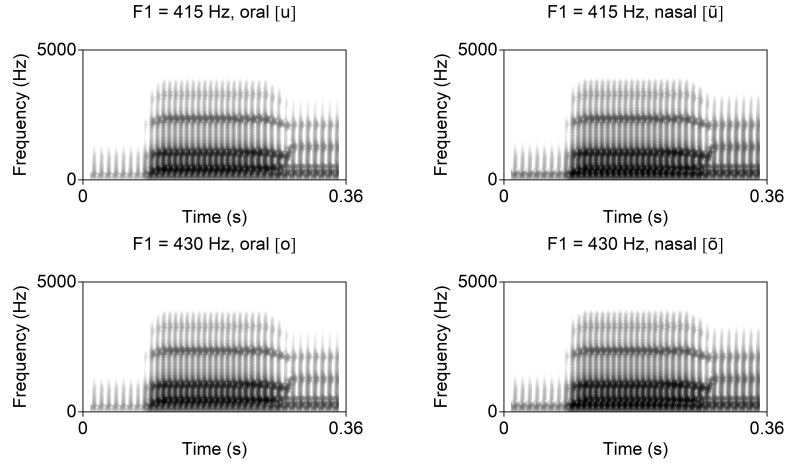


Figure 13: Spectrograms of the stimuli used in Experiment 2, which consisted of synthesized syllables of the shape [bVn].

was positively correlated both in the sense that F1 and N increase or decrease together and that (assuming that increasing nasality makes a vowel seem more nasal) these changes are expected to produce the same effect of perceived vowel lowering; the latter pair is negatively correlated in the same sense. Finally, the two selective attention tasks involved the categorization of all four stimuli on the basis of just one dimension: F1 ([u] and [ũ] vs. [o] and [õ]), or N ([u] and [o] vs. [ũ] and [õ]). This task required the participant to selectively attend to the dimension relevant for categorization while ignoring irrelevant variation on the other. The full set of tasks and the stimuli used in each is given in Table 5; the same information is visualized in Figure 14.

In terms of the proportion of responses correct for each task, if F1 and N are perceptually orthogonal then performance on all three tasks should be equivalent. In other words, the presence or absence of variation on one dimension should make no difference in judgments on the other. If F1 and N display perceptual integrality, then correlated performance should be better than baseline performance, which in turn should be better than selective attention perfor-

Task	Category 1	Category 2
Baseline 1	[u]	[o]
Baseline 2	[ũ]	[õ]
Baseline 3	[u]	[ũ]
Baseline 4	[o]	[õ]
Correlated 1	[ũ]	[o]
Correlated 2	[u]	[õ]
Selective attention 1	[u, ũ]	[o, õ]
Selective attention 2	[u, o]	[ũ, õ]

Table 5: List of tasks (conditions) and the stimuli used in each in Experiment 2. The labels ‘Yes’ and ‘No’ were randomly assigned to categories 1 and 2 for each participant and task.

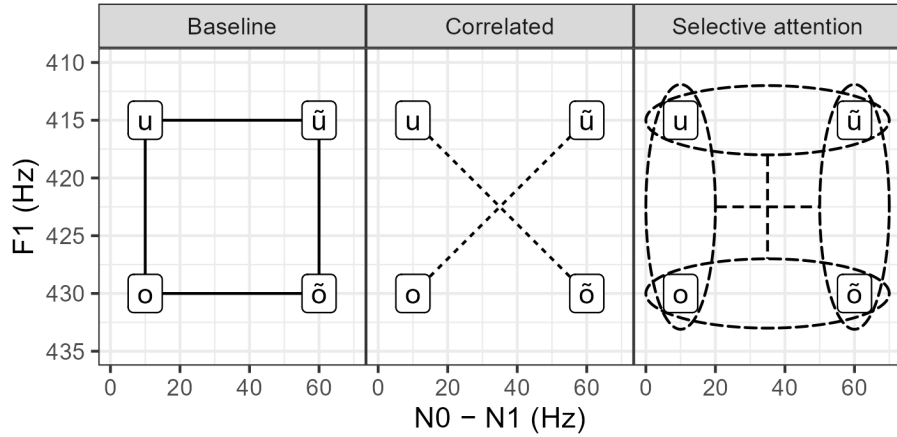


Figure 14: Arrangements of stimuli in each task. Each line represents a task and connects the (sets of) stimuli used in that task.

mance. This is predicted to occur because the perceived difference between stimuli which differ on two dimensions should be greater than the perceived difference between stimuli which vary on just one. The correlated task aligns categorization judgments with this greater difference, resulting in better performance. Meanwhile, the selective attention task requires that perceivers disregard information that they would normally use to make judgments because they cannot know whether it should be attributed to the dimension on which they must make judgments or to other dimension whose variation is irrelevant.

A more detailed explanation of this reasoning is given in §3.2.

3.1.3 Procedures

In each task, participants were instructed to label two categories of stimuli ‘Yes’ and ‘No’. The assignment of labels to categories varied arbitrarily between tasks to prevent participants from developing associations between labels and stimulus characteristics. Participants were told explicitly that the labels varied, and that they must pay careful attention to feedback to determine which label was used for which category in a given task.

Individual trials were structured as follows. Participants were seated at a computer in a quiet room wearing headphones. At the beginning of each trial, a fixation cross appeared for 500 ms, after which a 360 ms stimulus played through the headphones. After the presentation of the stimulus, the fixation cross was replaced with the labels ‘Yes’ over an ‘F’ and ‘No’ over a ‘J’. Participants had 1500 ms in which they could respond by pressing the ‘F’ key for ‘Yes’ and the ‘J’ key for ‘No’. Upon the detection of a keystroke, the response was logged and feedback given, displaying either ‘Correct’ or ‘Incorrect’. If no response was detected within 1500 ms, ‘Too slow!’ was displayed instead. Feedback messages were displayed for 1000 ms. The screen was then blanked for 500 ms, after which the next trial began. The length of a trial was thus between 1860 and 3360 ms, with an intertrial interval of 500 ms and an interstimulus interval of between 2000 and 3500 ms. The response window duration of 1500 ms was chosen as being long enough to avoid excessive time outs but short enough to prohibit conscious reasoning about how to respond. The other intervals were chosen arbitrarily.

The experiment consisted of a training phase followed by the main experiment. In both phases, each individual task involved 40 presentations of each stimulus, resulting in $2 \text{ stimuli} \times 40 \text{ repetitions} = 80 \text{ trials}$ in each baseline

and correlated task and $4 \text{ stimuli} \times 40 \text{ repetitions} = 160$ trials in each selective attention task.

In the training phase, participants categorized the stimuli by height and by nasalization in a series of three tasks ordered by increasing complexity: i) baseline F1 ([u] vs. [o]), ii) negatively correlated F1 and nasality ([ũ] vs. [o]), and iii) selective attention to F1, disregarding nasality (u) and [ũ] vs. [o] and [ō]). Each task was the same length as in the main experiment (80, 80, and 160 trials, respectively). The training blocks were not interspersed with the experimental blocks for three reasons. First, as mentioned in §3.1.2 it was important for performance to be below ceiling to facilitate the detection theoretic analysis discussed in §3.2.2. Training before each experimental block would have increased performance on those blocks, increasing the likelihood of ceiling effects. Second, placing the training before the main experiment allowed it to be used to screen participants who would likely perform at chance throughout the experiment and thus produce unusable data. To this end, participants had to score at least 55% correct on at least one task in the training phase to proceed to the main experiment. Only one participant did not reach this criterion. Finally, Experiment 2 was intended as a partial replication of Kingston and Macmillan’s (1995) Experiment I. Their experiment did not intersperse training and experimental blocks; doing so in Experiment 2 would have introduced an unnecessary difference from their design.

The main experiment was divided into eight blocks, each of which consisted of one task. With six blocks of 80 trials and two blocks of 160 trials, there were $6 \times 80 + 2 \times 160 = 800$ experimental trials in total, less than half as many as the 1924 trials of Kingston and Macmillan’s (1995) Experiment 1. This difference results from the fact that i) Kingston and Macmillan included an additional task designed to test a separate hypothesis and ii) they ran their

participants through the experiment twice. Despite this disparity in the total number of trials, the present experiment has a greater number of observations per condition per participant because there were only four stimuli presented 40 times in each task whereas Kingston and Macmillan’s vowel stimuli were presented in eight different [C_C] frames and in isolation 10 times each.

The order in which participants completed the blocks was varied according to a Latin square design. Participants were assigned block orders based on the order in which they participated. Participants were given a short break every 40 trials (roughly every 2–3 minutes) and a longer break between blocks. The experiment was divided into two sessions of four blocks each, plus training in the first session. The first session generally took 60–90 minutes to complete and the second 30–45 minutes.

3.2 Results

Two analyses are discussed in this section: a ‘classical’ Garnerian analysis, and a detection theoretic analysis modeled on the analysis give in Kingston and Macmillan (1995). The former uses proportion correct $p(c)$ as its measurement, and is discussed first. The latter uses perceptual distance d' as its measurement; the majority of the discussion focuses on this analysis. All data analysis was conducted in R.

3.2.1 Classical interpretation

The classical interpretation of $p(c)$ results in the Garner paradigm (Garner, 1974, ch. 7) is that roughly equivalent performance across tasks indicates no integrality, while differential performance in the ordering $p(c)_{correlated} > p(c)_{baseline} > p(c)_{selective}$ diagnoses integrality. The reasoning behind this is that integrality between two dimensions results in a perceived covariation that

may not actually exist. In information theoretic terms, this represents a loss of information: integrality results in information from two sources, e.g., F1 and nasality, becoming partially commingled with no way of knowing which source the information came from.

In the correlated tasks, because the actual physical covariation (once noticed) can be relied upon, the observer can treat it as redundancy in the signal and use accurate perception of one dimension to reliably make up for potentially deficient perception of the other ('redundancy gain'). For example, poor perception of F1 (perhaps limited to certain frequency ranges) might be made up for by good perception of nasality or vice versa. It does not matter where the information came from so long as it reliably allows the observer to categorize accurately. However, in the selective attention tasks, the commingled information is unreliable because some of it comes from the dimension of interest while some of it does not and may be misleading. The observer must filter out this unreliable information along with information they know to represent irrelevant variation, reducing performance ('filtering loss'). For example, a listener trying to categorize by height while nasalization is varied is forced to base height judgments on the subset of the signal which they can reliably attribute to F1 rather than N, which is smaller than F1's actual contribution.

To assess whether F1 and N displayed integrality, mean $p(c)$ values for the baseline, correlated, and selective attention tasks were calculated and a comparison between these values conducted to determine whether participants showed redundancy gains and/or filtering losses. These comparisons are visualized in Figure 15.

For five participants, $p(c)_{correlated} > p(c)_{baseline} > p(c)_{selective}$, i.e., these participants displayed both filtering gains and redundancy losses; these are the points occupying the area highlighted red in Figure 15. For four other

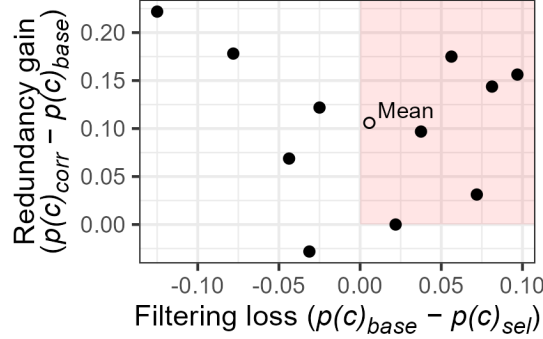


Figure 15: Scatter plot of redundancy gain ($p(c)_{corr(related)} - p(c)_{base(line)}$) compared to filtering loss ($p(c)_{baseline} - p(c)_{sel(ective)}$) for each participant (filled circles), plus mean values for all participants (empty circle). The origin represents $p(c)_{correlated} = p(c)_{baseline} = p(c)_{selective}$, i.e., no integrality. The quadrant highlighted in red represents $p(c)_{correlated} > p(c)_{baseline}$ and $p(c)_{baseline} > p(c)_{selective}$, i.e., simultaneous redundancy gains and filtering losses.

participants, $p(c)_{correlated} > p(c)_{baseline}$ but $p(c)_{baseline} < p(c)_{selective}$, i.e., they displayed performance gains in both the correlated and selective attention tasks relative to the baseline tasks. The two remaining participants did not show redundancy gains. Mean $p(c)_{correlated} = .76$ while mean $p(c)_{baseline} =$ mean $p(c)_{selective} = .65$, meaning that on average there was redundancy gain but not filtering loss. The mean $p(c)_{selective}$ of .65 is likely not meaningful because it reflects the average of two distinct groups of participants: those who performed worse in the selective attention tasks relative to the baseline tasks (i.e., who showed filtering losses) and those who performed better.

What factor divided these two groups of participants? One possible explanation is that for the group who showed filtering losses the degree of integrality between perceived F1 and nasality was relatively consistent across tasks, leading to redundancy gains and filtering losses as predicted. Meanwhile, for the group who did not show filtering losses, the degree of integrality was not consistent but could be increased beyond a certain base level in order to exploit redundancies

in the signal when they happened to be present and reliable, as in a block of 80 trials guaranteed to contain just two stimuli in which F1 and N covary. In other words, the group who did not show filtering losses might have actively increased the degree of F1-nasality integrality in their perceptions in order to seek out redundancy gains when they were confident that F1 and N would covary, and otherwise decreased it to minimize filtering losses.

3.2.2 Detection theoretic interpretation

The primary analytic approach used to interpret the results of Experiment 2 is a detection theoretic (Hautus, Macmillan, & Creelman, 2021) interpretation of the Garner paradigm modeled on Kingston and Macmillan’s (1995). It formalizes the logic previously presented to justify redundancy gain and filtering loss by modeling both as distortions of a decision space that increase the perceptual distance between some categories while decreasing the perceptual distance between others. Increases in the perceptual distance between stimuli translate into commensurately increased performance when categorizing those stimuli, and decreases in perceptual distance translate into decreased performance.

Detection theory models perceivers’ categorization on a given perceptual dimension as the division of that dimension into regions separated by decision boundaries, with stimuli categorized based on which region their perceived value falls into. For example, perceived nasality might be divided into an ‘oral’ and ‘nasal’ region, with stimuli being categorized as oral if their perceived nasality falls on that side of the boundary and as nasal otherwise. Additional dimensions beyond the first result in a decision *space*.

The perceived value of a stimulus on a given dimension does not have a deterministic relationship to the corresponding physical value(s). Even when the physical value is held constant, variation in the perceived value will be introduced by various sources of noise within the observer’s perceptual system, e.g.,

contrast effects between the spectra of adjacent sounds (Kingston et al., 2014), the conflation of perceived duration and frequency (Brugos & Barnes, 2012), or simple electrical noise in the nervous system. Consequently, perceived values are typically modeled as normally distributed on the perceptual dimension in question. A 2-dimensional decision space with four bivariate distributions corresponding to the stimuli in Experiment 2 is visualized in Figure 16.

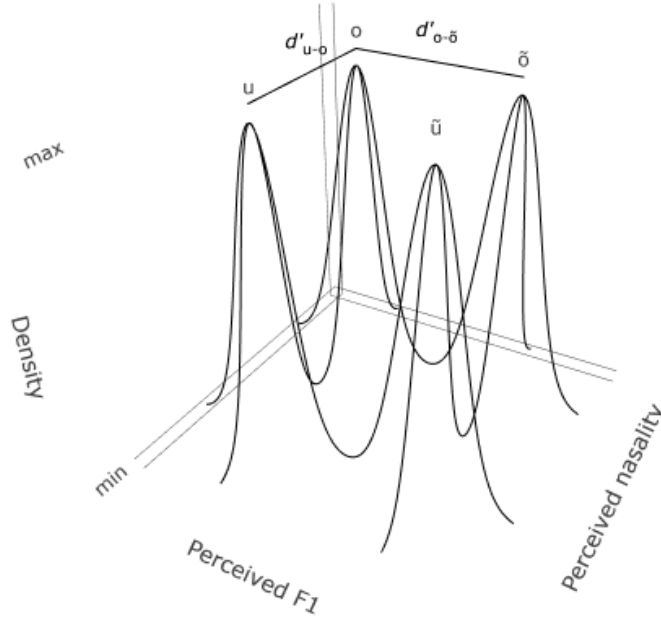


Figure 16: Hypothetical perceived F1 and nasality distributions for the stimuli used in Experiment 2 in the 2-dimensional decision space of the perceived F1-perceived nasality plane. Euclidean distances d' between distribution means are labeled by the distributions they separate. Decision boundaries are not shown.

The Euclidean distance d' between the means of a given pair of distributions is directly proportional to performance $p(c)$ in assigning stimuli to one of those two responses. As the d' between two means approaches 0, the distributions of perceived values arising from the corresponding physical inputs overlap more and more, resulting in worse categorization performance until they finally merge

completely and performance drops to chance, i.e., $p(c)$ goes to .5.

Conversely, as d' approaches infinity $p(c)$ approaches 1 and vice versa. $d' = \infty$ is not a meaningful statement because perceptual distance cannot be literally infinite. To avoid infinite d' , when a participant answered ‘Yes’ in every trial in which ‘Yes’ was the correct answer, the resulting hit rate of 1 was converted to $1 - 1/(2N)$ where N is the number of trials, e.g., $1 - 1/(2 \times 40) = .975$ (Macmillan & Kaplan, 1985; Murdock & Ogilvie, 1968, as cited in Hautus, Macmillan, & Creelman, 2021). This correction was applied to two blocks of trials out of 88; in both cases the task was Correlated 1 (negative correlation).

Integrality is modeled as the non-perpendicularity of the two perceived dimensions in decision space. When the decision space depicted in Figure 16 is viewed from overhead (i.e., looking directly onto the perceived F1-perceived nasality plane), the four means form a parallelogram. As a simplifying assumption, only parallelograms (arrangements of means in which opposing sides are equal in length and diagonally opposing corners form equal angles) are considered. Figure 17 shows two such parallelograms, one in which perceived F1 and nasality are completely separable (left) and one in which they are integral (right). In the separable parallelogram, the perceived nasality and F1 axes are perpendicular to each other, so a change in the perceived value of a stimulus on one axis does not imply any change in that stimulus’ perceived value on the other axis and vice versa. In the integral parallelogram, the axes are not perpendicular to one another, meaning that a change in the perceived value of a stimulus on one axis does imply a change on its perceived value on the other axis.

The estimated internal angle θ at the corners connected by the negatively correlated diagonal $d'_{\bar{u}-o}$ is the statistic which summarizes the extent to which the two perceptual dimensions are integral. $\theta = 90$ corresponds to complete

separability of perceived F1 and nasality. It also entails that the two diagonal d 's corresponding to categorization performance in the correlated tasks will be equal. $\theta < 90$ corresponds to the (partial) collapse of perceived F1 and nasality such that $d_{\tilde{u}-o} > d_{u-\tilde{o}}$; $\theta > 90$ corresponds to the (partial) collapse of perceived F1 and nasality such that $d_{u-\tilde{o}} > d_{\tilde{u}-o}$.

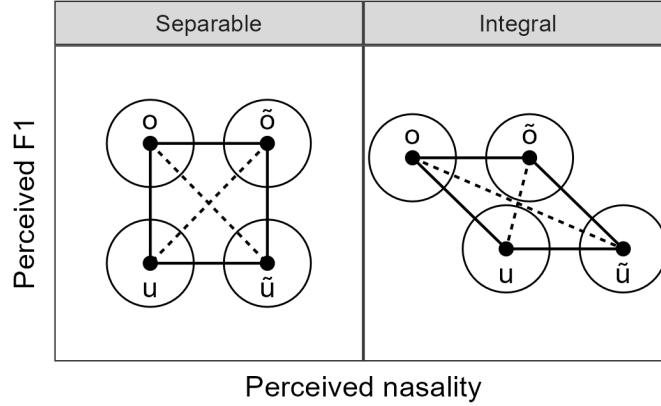


Figure 17: Overhead view of decision spaces with separable and integral dimensions. Points represent means, circles contours of equal density in the distributions around the means, solid lines d 's between stimuli differentiated by either F1 or nasality as in the baseline tasks, and dashed lines d 's between stimuli differentiated by both F1 and nasality as in the correlated tasks. The value of the angle θ at $[o]$ is the measure of integrality.

The interpretation of the Garner paradigm results differs between the detection theoretic and classical approaches. Rather than performance differences corresponding directly to the type of task, the detection theoretic approach predicts a ‘redundancy gain’ (performance increase relative to the baseline tasks) for any correlated task in which d' is greater than the baseline d 's and a ‘filtering loss’ for any correlated task in which d' is smaller than the baseline d 's. This leads to two major differing predictions.

First, redundancy gains alone do not diagnose integrality. As θ approaches 0° the positively correlated $d_{u-\tilde{o}}$ shrinks and the negatively correlated $d_{\tilde{u}-o}$

grows; as θ approaches 180° , the opposite occurs. When $\theta = 60^\circ$, $d_{\text{u-}\tilde{\text{o}}}$ is equal to one of the baseline d 's; at $\theta = 120^\circ$, $d_{\text{u-o}}$ is equal to one of the baseline d 's. Only when $\theta = 90^\circ$ (when the dimensions are separable) are the correlated d 's also equal to each other. Consequently, it is unequal performance in the two correlated tasks that diagnoses integrality, not just greater performance in the correlated tasks than in the baseline tasks.

Second, redundancy gains in both correlated tasks are expected when $60^\circ < \theta < 120^\circ$ as both correlated d 's are greater than the baseline d 's, but redundancy gains in one correlated task and filtering losses in the other are expected when $\theta < 60^\circ$ or $\theta > 120^\circ$ as one of the correlated d 's will be smaller than one or both of the baseline d 's. In other words, as the degree of integrality increases, performance in one correlated task gets better while performance in the other worsens until it turns from a redundancy gain to a filtering loss.

Both $p(c)$ and d' were calculated for each participant in each of the eight tasks. The results for the two-stimulus (baseline and correlated) tasks are reported in Table 6.

Task	Categories	$p(c)$ (SD)	d' (SD)
Baseline 1	[u], [o]	.61 (.10)	0.58 (.54)
Baseline 2	[ũ], [õ]	.58 (.10)	0.42 (0.54)
Baseline F1		.59 (.10)	0.50 (0.53)
Baseline 3	[u], [ũ]	.71 (.14)	1.35 (1.11)
Baseline 4	[o], [õ]	.71 (.14)	1.93 (1.02)
Baseline N		.71 (.14)	1.31 (1.04)
Correlated 1	[ũ], [o]	.78 (.16)	1.93 (1.43)
Correlated 2	[u], [õ]	.73 (.16)	1.47 (1.12)
Correlated (all)		.76 (.16)	1.70 (1.28)
Selective attention 1	[u, ũ], [o, õ]	.58 (.09)	
Selective attention 2	[u, o], [ũ, õ]	.71 (.09)	
Selective attention (all)		.65 (.11)	

Table 6: Mean $p(c)$ and d' with standard deviations across participants in Experiment 2. d' is not reported for the selective attention conditions because it is a measure of perceptual distance between two stimuli, not four.

In the baseline tasks, performance was much better when categorizing by nasality compared to F1, particularly for the stimulus pair [o, $\tilde{o}$]; in the F1 baseline tasks average $d' = 0.50$, while in the baseline nasality tasks average $d' = 1.31$. Given that Kingston and Macmillan (1995) found a similar result and that the stimuli used in Experiment 2 were based on theirs, it is likely that categorization performance on the basis of nasality was superior for the same reason that Kingston and Macmillan give: the F1 difference in the stimuli was close to 1 JND while the N difference likely exceeded 1 JND. Many participants reported that some of the stimuli seemed louder than others; it is possible that although the stimuli were normalized to an intensity of 70 dB, the greater low-frequency content of the nasalized stimuli was relatively easy to detect as perceived greater intensity. In other words, the effect was likely specific to these stimuli and not a result of nasality being inherently easier to judge than F1.

Performance in the correlated tasks was not equal. In the positively correlated task ([u], [$\tilde{o}$]) $p(c) = .73$; in the negatively correlated task ([$\tilde{u}$], [o]) $p(c) = .78$. This indicates that perceived F1 and perceived nasality were integral such that $0^\circ \leq \theta \leq 90^\circ$.

θ was estimated for each participant using the following procedure. For each participant, d'_{u-o} was averaged with $d'_{\tilde{u}-\tilde{o}}$ to yield d'_{F1} , and $d'_{u-\tilde{u}}$ with $d'_{o-\tilde{o}}$ to yield d'_N . Parallelograms with height d'_{F1} and width d'_N and θ ranging from 0° to 180° were constructed. The ratio between the diagonals of each parallelogram was compared to the ratio between the positive and negative correlated d' s, and the best fit parallelogram was selected. The resulting parallelograms are shown in Figure 18.

Across all participants, mean $\theta = 67^\circ$, i.e., perceived F1 and nasality were modestly integral such that $d'_{\tilde{u}-o} > d'_{u-\tilde{o}}$, i.e., categorization performance in the correlated task with negatively covarying F1 and nasality was better. This

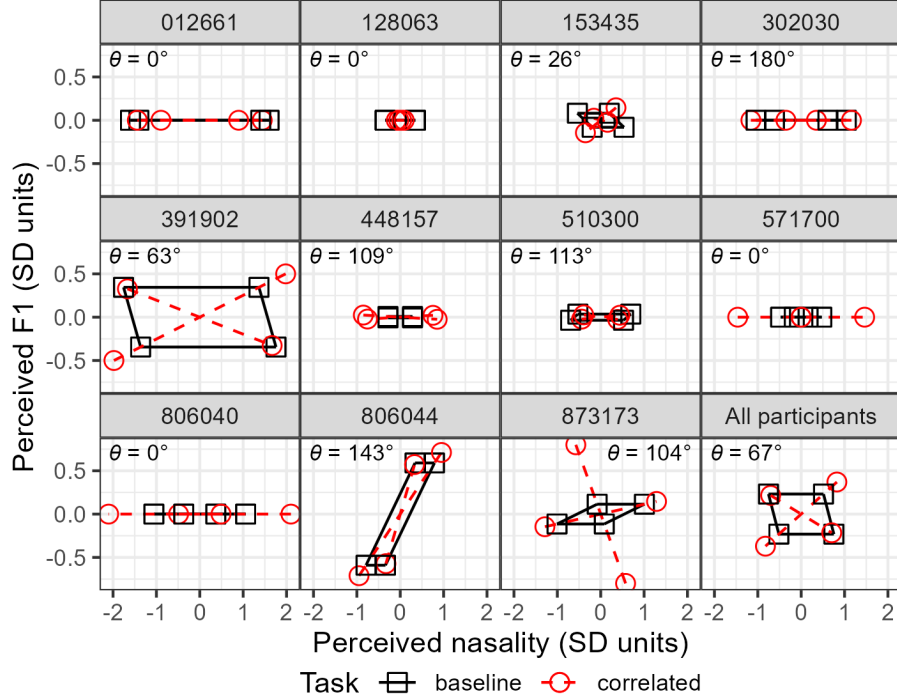


Figure 18: Decision spaces with parallelogram representations of data for each participant, plus an average decision space for all participants. Lines represent d' values; points represent average locations of means.

effect is similar in direction but not size to results reported by Kingston and Macmillan (1995), who found $\theta = 61^\circ$ for vowels in isolation but $\theta = 23^\circ$ in the frames [b_n] and [d_n], which are more comparable to the stimuli used in Experiment 2.

There was considerable between-participant variation not seen in Kingston and Macmillan (1995). For four participants $\theta = 0^\circ$ indicating complete integrality of perceived F1 and nasality such that $d'_{\bar{u}-o} > d'_{\bar{u}-\bar{o}}$, i.e., negative covariation of F1 and N facilitated categorization. For one participant $\theta = 180^\circ$, indicating complete integrality in the opposite direction: *positive* covariation of F1 and N facilitated categorization. The remaining participants showed degrees

of integrality ranging from 26° to 143° , with a cluster of three at 104° , 109° , and 113° .

A leave-one-out cross-validation (LOOCV) analysis was conducted to better understand how divergent the θ values were from each other. Each participant's θ was compared to a mean calculated from the other participants' θ s. The RMS error of this set of comparisons was 69.13° or 38% of the range between 0° and 180° . This extremely large error value strongly suggests that the current data set is too small to draw summary conclusions about integrality between perceived F1 and nasality across any population of native English speakers. In other words, the between-participant mean of $\theta = 67^\circ$ is likely not meaningful, nor are any statistics derived from it.

Kingston and Macmillan (1995) argue that for any given parallelogram it is possible to predict the degree of redundancy gain and filtering loss as a function of the relevant d' values. As a second form of cross-validation and additional point of comparison with their study, observed performance in the correlated and selective attention tasks was compared with performance predicted from the baseline tasks and θ . A parallelogram was calculated for each participant using their observed d'_{F1} , d'_N , and θ values, and the correlated d' s (diagonals) calculated. d' is a measure of perceptual distance between a pair of stimuli meaning that it cannot be applied to a task with a greater number of stimuli such as the selective attention tasks. Consequently, the relevant d' values were converted to $p(c)_{max}$ ($= \Phi(d'/2)$), the maximum possible $p(c)$ for a given d' assuming no decision bias towards one stimulus. In the correlated tasks, the relevant d' was one of the parallelogram diagonals; in the selective attention tasks, the relevant d' s were all but those between stimuli in the same category, i.e., all but the two d'_N s in the F1 task and all but the two d'_{F1} s in the nasality task. For the selective attention tasks, the four resulting $p(c)_{max}$ values were

averaged to produce a final predicted $p(c)_{max}$. For each participant in each task, the predicted $p(c)_{max}$ values were compared to the observed values. The results of this comparison are reported in Table 7.

Task	$p(c)_{max}$ (pred.)	$p(c)$ (obs.)	Difference	RMSE
Corr. 1 (neg.)	.75 (.12)	.78 (.16)	-.03 (.11)	.11
Corr. 2 (pos.)	.74 (.14)	.73 (.16)	.01 (.09)	.08
Sel. attn. 1 (F1)	.67 (.08)	.58 (.09)	.09 (.10)	.13
Sel. attn. 2 (N)	.73 (.12)	.71 (.09)	.02 (.08)	.08

Table 7: Mean predicted $p(c)_{max}$, mean observed $p(c)$, the difference, and root mean square error (RMSE) of the differences for the non-baseline tasks. Standard deviations are given in parentheses.

The difference between observed and predicted values was 3% or less with the exception of the F1 selective attention task, which had a difference of 9% possibly due to its lower $p(c)$ compared to the other tasks. While this is suggestive of good congruence between the results of the baseline and non-baseline tasks within the detection theoretic model, the standard deviations and root mean square error (RMSE) were larger than the effect sizes, indicating that these estimates are likely unreliable.

Another test of congruency between the results of the baseline and correlated tasks was conducted by comparing estimated correlated d' values against observed ones. Correlated d' values were estimated from the baseline data and θ values using the method described for the $p(c)_{max}$ cross-validation and compared with observed correlated d' values. The difference between mean estimated and observed d'_{u-o} was 0.42 (SD: 0.95; RMSE: 0.65) or 10%; the difference between mean estimated and observed $d'_{\bar{u}-o}$ was 0.07 (SD: 0.62; RMSE: 0.26) or 2%. This indicates that there was substantial variation in the degree of congruence between the decision spaces implied by the baseline and correlated tasks. As with the $p(c)_{max}$ cross-validation, the standard deviations and RMS errors are larger than the effect sizes, indicating that these estimates are likely unreliable.

Every participant showed at least moderate integrality between perceived F1 and nasality, and five showed complete integrality. There was substantial between-participant variation with θ ranging from 0° to $\theta = 180^\circ$, meaning that the all-participant mean of $\theta = 67^\circ$ is likely not meaningful. Because each participant contributes only a single data point (their θ value), with only 11 participants there was too little data for the variation in θ to resolve into clear trends. The fact that Experiment 2 was underpowered was further supported by large error values in both cross-validations. Overall, it is unlikely that the data from Experiment 2 by themselves can be meaningfully interpreted.

3.3 Production-perception correlation

Correlation between covariance of F1 and nasalance in production and integrality in perception was calculated using Spearman’s rank correlation test, both for consistency of statistical method and because the participants’ θ values are not normally distributed. The F1-nasalance ρ_s values reported in §2.2 were used to represent F1-nasalance covariation in production. Four correlations were calculated using F1-nasalance ρ_s values calculated across all words and by word type. To avoid confusion, ρ_{sP} refers to F1-nasalance correlations in the production data while ρ_{sI} refers to correlations between ρ_{sP} and θ .

While each correlation test yielded very large correlations, none reached significance ($p < .05$). The correlations were as follows: θ with ρ_{sP} (all words), $\rho_{sI} = .59$, CI $[-.04, .88]$; θ with ρ_{sP} (fillers), $\rho_{sI} = .51$, CI $[-.15, .86]$; θ with ρ_{sP} (loners), $\rho_{sI} = .53$, CI $[-.13, .86]$; and θ with ρ_{sP} (quartet words), $\rho_{sI} = .32$, CI $[-.37, .78]$. Additionally, the difference between loner and quartet ρ_{sP} values showed a very large but non-significant correlation with θ , $\rho_{sI} = .49$, $p \geq .05$, CI $[-.17, .85]$. These statistics are visualized in Figure 19.

While these correlations appear suggestive of a relationship between integral-

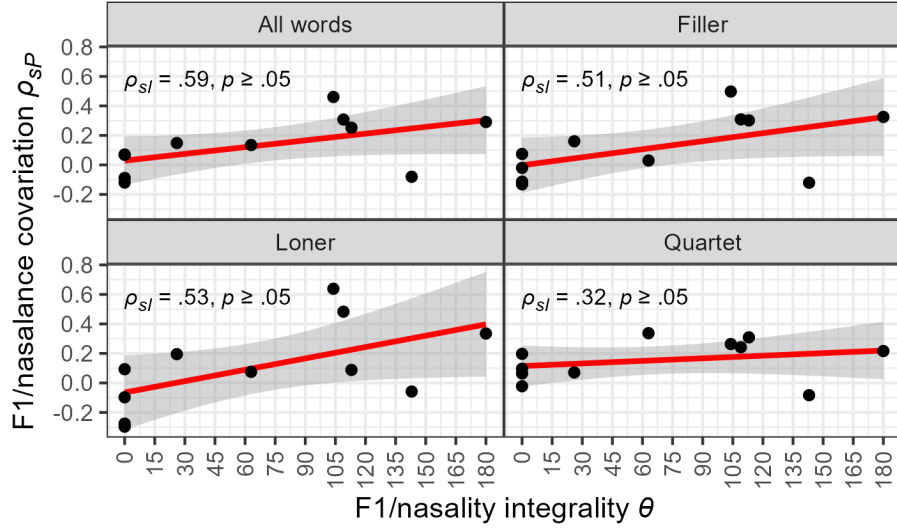


Figure 19: Scatter plots of ρ_{sP} and θ , plus best-fit correlation lines. Shaded areas represent a 95% confidence interval.

ity in perception and covariation in production as argued for by Kingston and Diehl (1994), their extremely wide confidence intervals of 50% or more indicate that they are highly unreliable. An additional LOOCV analysis was conducted to understand whether this unreliability was the result of a small number of outliers or a property of the entire data set. Each participant's θ was rescaled to the same range as ρ_{sP} , $[-1, 1]$, and compared against four different means calculated from the other participants' ρ_{sP} values for all words, fillers, loners, and quartet words, respectively. The RMS error of the comparisons were .73 or 37% for all word ρ_{sP} , .71 or 36% for filler ρ_{sP} , .69 or 35% for loner ρ_{sP} , and .79 or 40% for quartet word ρ_{sP} . This indicates that many θ values diverged substantially from what would be expected if θ and ρ_{sP} were linearly related and that the unreliability of the correlations was not due to the influence of a small number of outliers.

3.4 Discussion

An experiment in the Garner paradigm was conducted to estimate each participant’s degree of integrality between perceived F1 and nasality. The stimuli were four synthesized vowels which were varied orthogonally in F1 and N and placed in a [b_n] frame. Participants categorized the stimuli in a series of 2AFC tasks which established the perceptual distance d' between each pair of stimuli. Based on these d' values, a decision space represented as a parallelogram with the stimuli at the four corners was calculated for each participant. The internal angle θ at the corner corresponding to the oral non-high stimulus was the measure of integrality.

There was substantial between-participant variation in the θ values obtained in Experiment 2, and it was determined that the mean θ of 67° was unlikely to be meaningful. It is possible that the between-participant variation arose because participants approached the 2AFC task in different ways at different times, treating the stimuli variously as words, as syllables, or as non-linguistic sounds. Indeed, many of the participants reported as much in their debriefing questionnaires. The consequence is that Experiment 2 recorded a wide range of θ values reflecting these different approaches rather than a narrower range of values reflecting the language-specific behavior which is the object of this investigation.

One consequence of the between-participant variation in θ values is that the integrality between perceived F1 and nasality observed in Experiment 2 must be at least partially learned and not inherent to the human perceptual system. If the degree of F1-nasality integrality were determined by some factor in human physiology or neurology such as the functioning of the auditory system or the processing of auditory input in the brain, then it should be similar across all humans. However, observed θ was distributed across the entire range of

possible values from 0° to 180° . This finding appears irreconcilable with any model of integrality that does not include learning as a means of introducing between-listener variation. Additionally, there is to the best of my knowledge no proposed physiological mechanism which would cause perceived F1 and nasality to be integral. Until one is identified, the only arguments in favor of totally inherent specification of integrality are the related preferences for theoretical parsimony and for variation to be reducible to an underlyingly invariant factor.

If the degree of integrality between perceived F1 and nasality is learned, what factors might condition its variation? There is ample evidence that the perceptual criteria by which native listeners categorize speech sounds are malleable, with their adjustment being triggered by a wide variety of factors. For example, category boundaries are sensitive to the phonetic context: native English listeners were more likely to judge ambiguous sounds from a [s]-[ʃ] continuum as [s] before a rounded vowel [u] and as [ʃ] before an unrounded vowel [a], likely because before [u] rounding can be attributed to the vowel as well as the fricative (Mann & Repp, 1980). They are also sensitive to lexical regularities, e.g., native English listeners were more likely to judge ambiguous sounds from a [ɹ]-[l] continuum as [ɹ] in the context [t__i] and as [l] in the context [s__i] because [tɹi] and [sli] are common sequences in English but [tli] and [sɹi] are (nearly) unattested (Massaro & Cohen, 1983). Speaker idiosyncrasies also affect category boundaries. Clarke and Luce (2005) exposed native English listeners to modified English speech in which onset /d/ and /t/ were realized with reduced voice onset times (VOTs). After hearing as few as 20 stops with non-canonical VOTs, listeners' categorization boundaries on a [d]-[t] stimulus continuum shifted towards [d], i.e., they categorized more stops with short VOTs as [t], reflecting the modified speech. These are just a few of the many factors which affect listeners' categorization of speech sounds; for a review, see Cutler (2012, ch. 11

and citations therein).

It is not difficult to imagine that the degree of integrality could show similar sensitivity to phonetic context, lexical regularities, and speaker idiosyncrasies, among other factors. Indeed, Kingston and Macmillan (1995) found that the value of θ for perceived F1 and nasality in vowels was sensitive to segmental context. The vowels in their stimuli could be followed by [b], [d], [m], or [n]; θ was equal to 44° for vowels before [d], 25° before [m], 23° before [n], and 0° before [b]. If the degree of integrality can vary with segmental context, we must also consider the possibility that it varies with lexical regularities and/or speaker idiosyncrasies.

While the influence of lexical factors on integrality was not tested in Experiment 2, the exposure of participants to sets of stimuli showing different patterns of F1-N covariation in different tasks of the Garner paradigm may be analogized to listeners' exposure to speakers with different degrees of F1-nasality covariation, or the same speaker producing groups of lexical items showing different degrees of F1-nasality covariation (as was observed in Experiment 1). As noted in §3.2.2, there were in fact discrepancies between the decision spaces implied by data from the baseline and correlated tasks. I speculate that listeners may have adjusted the degree of F1-nasality integrality in their perceptions in order to take advantage of the covariation of F1 and N in the correlated tasks' stimuli, analogous to Clarke and Luce's (2005) listeners adjusting their [d]-[t] categorization boundaries after exposure to [d] and [t] tokens with non-canonical VOTs. A change in the degree of integrality between tasks would also imply that the it can be adjusted in real time or at least very rapidly in response to the task at hand. Speculating further, between-participant variation in the size of the difference may indicate that some participants adjusted the degree of integrality across a wider range of values than other participants did, i.e., listeners showed

different degrees of flexibility in how much they could adjust their degree of integrality. This would suggest that not only is integrality learned, but so is the degree to which it can be adjusted.

The possibility that the degree of integrality can vary dynamically as a function of the task also raises a potential issue with the detection theoretic interpretation of the Garner paradigm. As described in §3.2.2, the estimation of θ relies on the comparison of d' values observed in the correlated tasks with correlated d' values estimated based on the baseline data. This comparison assumes that the decision space is consistent across tasks. If the degree of integrality can change between tasks, then that assumption fails and θ cannot be reliably estimated. As an extreme example, suppose that a listener adjusts the degree of integrality such that $\theta = 0^\circ$ in the negatively correlated task and $\theta = 180^\circ$ in the positively correlated task. The two correlated d' s would be equal and the best-fit parallelogram estimated from the baseline data would therefore be a rectangle with equal diagonals, yielding an estimated $\theta = 90^\circ$ even though there was complete integrality in both correlated tasks.

While this appears to be a serious problem for the detection theoretic interpretation of the Garner paradigm, it should be possible to detect whether the degree of integrality has changed between tasks by observing the size of the difference between the decision spaces implied by the baseline and correlated data. If the difference is small, it is likely that the degree of integrality was similar between tasks. If the difference is large and especially if both observed correlated d' values are larger than their estimated counterparts, the degree of integrality may have changed between tasks. Both of these outcomes were observed in the data obtained from Experiment 2, suggesting that some participants may have maintained a consistent degree of integrality while others adjusted their degree of integrality between tasks.

The classical interpretation of the Garner paradigm may also be able to detect changes in the degree of integrality between tasks. In addition to adjusting the degree of integrality in the correlated tasks to exploit the covariation between dimensions and increase the perceptual distance between stimuli, listeners may also adjust the degree of integrality such that $\theta \approx 90^\circ$ in the selective attention tasks in order to minimize filtering losses. If this occurs, then redundancy gains should be observed in the correlated tasks without corresponding filtering losses in the selective attention tasks, i.e., $p(c)_{\text{correlated}} > p(c)_{\text{baseline}} \approx p(c)_{\text{selective}}$. This was the pattern observed for four participants in Experiment 2.

While the results of Experiment 2 in both the classical and detection theoretic interpretations of the Garner paradigm may suggest that listeners dynamically adjusted the degree of integrality between perceived F1 and nasality between tasks, these results are to be interpreted cautiously. Just four participants appeared to adjust their degree of integrality. This is a very small sample, meaning that any conclusions drawn from it are tentative at best. Further, the classical and detection theoretic interpretations do not completely agree on which participants showed evidence of adjusting their degree of integrality. Of the four participants who showed performance gains in both the correlated and selective attention tasks relative to the baseline tasks, three showed large discrepancies between the decision spaces implied by the baseline and correlated data, but one showed only a small discrepancy. Consequently, the arguments presented here in favor of the degree of integrality being adjustable should be regarded as the speculative construction of a hypothesis which remains to be tested in future experiments.

4 General discussion

The purpose of this study was to test Kingston and Diehl’s (1994) hypothesis that integrality between two phonetic properties in perception may lead language users to covary them in their productions so as to exploit the integrality and increase the perceptual distance between contrasting categories which differ in those properties. To that end, two experiments were conducted to measure the degree of F1-nasalance covariation in production (Experiment 1) and the degree of F1-nasality integrality in perception (Experiment 2) in the same population of participants and conduct a comparison of these two values. While a direct comparison could only be conducted within individuals because this was the most granular level of abstraction shared by both experiments, it is possible to narrow the hypothesis space on the basis of data from the individual experiments.

From Experiment 1, we can conclude that there is F1-nasalance covariation in production, and the size of this covariation varies substantially between speakers and across lexical items. Specifically, the covariation is smaller for words which are part of minimal pairs involving height and/or the nasality of the postvocalic consonant. Overall, it seems that covariation between F1 and nasality must be described at the level of individual lexical items, or at least certain types of phonological neighborhoods such as those containing minimal pairs distinguished (in part) by the phonetic properties in question. From Experiment 2, we can conclude only that perceived F1 and nasality are integral among native English listeners and that there is substantial between-listener variation in the degree of integrality.

The data from Experiments 1 and 2 suggest but do not demonstrate a correlation between F1-nasalance covariation in production and F1-nasality integrality in perception. Measures of correlation between θ and F1-nasalance

covariation ρ_s were large but failed to reach significance. This may have occurred because there were so few θ values and because these values showed a large amount of variation. Consequently, it is not possible to demonstrate the existence of a production-perception correlation based solely on the data from Experiments 1 and 2.

However, the question to ask is not ‘Is there a correlation?’ but rather ‘At what level of abstraction might the correlation exist?’ The hypothesis space implied by Kingston and Diehl’s (1994) theory is large to the point that the enhancement hypothesis is difficult to conclusively falsify precisely because a production-perception correlation indicating the enhancement of contrasts (‘enhancement correlation’ for short) could exist at any of many different levels of abstraction, both in terms of group, individual, or within-individual analysis and in terms of units of contrast. This study directly tested for the existence of a correlation at just two levels of abstraction: across all data within a group (permitting no variation), and across all data for each individual (permitting between-individual variation). From this perspective, the scope of this study is tightly limited to the two most minimally specified versions of the enhancement hypothesis.

Other versions of the hypothesis that might be tested in future studies involve testing for production-perception correlations that exist i) within individuals and ii) within units of contrast smaller than ‘all vowels in the language’. Testing the enhancement hypothesis within individuals and at the level of units typically used to characterize language-wide contrasts, e.g., pairs of phonemes such as /ɪ/ and /ε/ within individuals, could be accomplished by establishing F1-nasality θ values for both phonemes for each participant and comparing these values to F1-nasalance covariation in all instances of those vowels produced by each individual. Testing the enhancement hypothesis within individuals and

at the level of lexical items would require establishing F1-nasality θ values for individual words.

Testing the enhancement hypothesis at finer levels of granularity will also require establishing whether the degree of integrality is in fact context-sensitive. For example, is the degree of integrality between perceived F1 and nasality the same in the vowels of *mist*, *mint*, *messed*, and *meant*? What about in the vowel *glitch*, which has no neighbors distinguished by vowel height or the nasality of the postvocalic consonant? If the F1-nasalance covariations found in Experiment 1 can be straightforwardly translated into θ values, then we should expect θ to be smaller for words like *mist*, which have neighbors distinguished by F1 and nasality and larger for words like *glitch*, which do not.

Setting aside the fact that only the two most minimally specified versions of the enhancement hypothesis were directly tested, which parts of the hypothesis space can we at least speculate are ruled out on the basis of the data from Experiments 1 and 2? The data show that covariation in production and integrality in perception both display between-participant variation, meaning that any production-perception link must also show variation between individual language users. In other words, there is not a single production-perception correlation which is consistent across the entire English language or the local speech community. The data from Experiment 1 also show that covariation in production varies between categories of words, including words with the same vowel in similar phonological contexts, meaning that there is not a single production-perception correlation at the level of the phoneme or allophone within individuals. If there is a relationship between covariation of phonetic properties in production and their integrality in perception, the relationship most likely exists at a finer level of granularity, i.e., within individuals and at the level of lexical items or the types of phonological neighborhoods those lexical items occupy.

A major reason why it is difficult to draw conclusive inferences from the results of Experiments 1 and 2 is that they were conducted at mismatching levels of abstraction. Experiment 1 measured F1-nasalance covariation at the level of the word; Experiment 2 measured F1-nasality integrality at the level of either the syllable, phoneme, or allophone (and for some trials it may not have measured integrality between F1 and nasality at all). In other words, Experiment 2 was designed with the assumption that integrality would be consistent across lexical items, while Experiment 1 did not make the same assumption about F1-nasalance covariation in production. These differing assumptions also led to mismatches in the amount of data that was ultimately generated. By the most conservative method of counting, each participant in Experiment 1 produced 80 data points (unique words); in Experiment 2 they produced just one (the θ value). This prevented a more detailed comparison of covariance in production with integrality in perception which might have revealed whether a relationship does exist at the level of the lexical item.

Because Experiment 2 was underpowered and the correlation of results from Experiments 1 and 2 was unreliable, it is not possible to make conclusive statements about which level of abstraction the production-perception correlation is likely to exist at. However, certain avenues of future investigation appear promising. The data are compatible with production-perception correlations at the level of the lexical item or its neighborhood, and Experiment 1 found variation in the size of F1-nasalance covariation between word neighborhood types. Future investigations might focus on assessing integrality at the same level of abstraction, i.e., the word and its neighborhood. For example, the paradigm used in Experiment 2 might be altered to make use of (synthesized) real words rather than synthetic vowels or syllables and incorporate a method for forcing participants to perform lexical access, preventing the non-linguistic approaches

that participants reported in Experiment 2.

Another future area of investigation might involve production-perception correlation at the level of the information-theoretic context, which was not investigated in this study. In combination with the results of the current study, Scarborough and Zellou’s (2013) finding that the magnitude of nasal coarticulation varies across words of different neighborhood densities and across different (simulated) speech contexts raises the possibility that the information-theoretic approach of Hypo- and Hyperarticulation Theory (Lindblom, 1990) could be extended to perception. In this view, listeners might modulate F1-nasalance integrality (and integralities between other phonetic properties) according to how much information they expect their interlocutor to transmit in the form of F1-nasalance covariation.

5 Conclusion

This paper tested a version of the hypothesis that integrality between phonetic properties in perception may lead to their covariation in production and consequently the enhancement of contrasts between categories differing in those properties (Kingston & Diehl, 1994). The discussion of this test was framed in terms of investigating the production-perception interface at multiple levels of abstraction simultaneously (Schertz & Clare, 2020), though ultimately the enhancement hypothesis was directly tested at just two levels of abstraction (the group and the individual speaker).

While the conclusion of the test must be the enhancement hypothesis’ falsification at the level of whole languages or individual speakers and likely also at the level of individual phonemes within speakers, the possibility of enhancement at greater levels of granularity such as individual lexical items is left open. Investigating this possibility will require the assessment of covariation of phonetic

properties in production and their integrality in perception at the same level of abstraction, ideally, the word.

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Appendices

A Experiment 1 items

V_h	V_l	Quartet words				Loners	
i	ɪ	seep	seem	sip	sim	gleeful	fixate
i	ɪ	skied	scheme	skid	skim	leafy	twinge
i	ɪ	weaker	wicker	wiener	winner	sequel	risky
i	ɪ	bead	bean	bid	bin	cuisine	symptom
i	ɪ	greet	green	grit	grin	teenage	fifty
i	ɪ	deep	deem	dip	dim	achieve	ginger
i	ɪ	sheep	sheen	ship	shin	senior	exist
i	ɪ	seek	scene	sick	sin	machine	image
ɪ	ɛ	hick	hymn	heck	hem	glitch	skeptic
ɪ	ɛ	mist	mint	messed	meant	sprinkle	festive
ɪ	ɛ	tick	tin	tech	ten	tricky	extend
ɪ	ɛ	lift	lint	left	lent	limbo	theft
ɪ	ɛ	pit	pin	pet	pen	distance	length
ɪ	ɛ	sits	since	sets	sense	instinct	friendly
ɪ	ɛ	did	din	dead	den	pretty	success
ɪ	ɛ	wit	win	wet	when	finished	every
ɪ	æ	grid	grim	grad	gram	citrus	rancid
ɪ	æ	spit	spin	spat	span	kindling	banjo
ɪ	æ	rip	rim	rap	ram	liquid	candid
ɪ	æ	chip	chimp	chap	champ	syringe	gamble
ɪ	æ	lip	limp	lap	lamp	vision	factor
ɪ	æ	trip	trim	trap	tram	income	passage
ɪ	æ	slip	slim	slap	slam	issue	sandwich
ɪ	æ	skit	skin	scat	scan	single	after
u	ʌ	duke	dune	duck	dun	duplex	deluxe
u	ʌ	ghoul	goon	gull	gun	gruesome	puncture
u	ʌ	booze	boon	buzz	bun	moonlight	shovel
u	ʌ	tube	toon	tub	ton	pursuit	amongst
u	ʌ	newt	noon	nut	nun	platoon	dozen
u	ʌ	root	rune	rut	run	improve	someday
u	ʌ	dude	doom	dud	dumb	assume	other
u	ʌ	rube	room	rub	rum	human	hundred
ʌ	ɑ	buck	bunk	balk	bonk	stubby	conquest
ʌ	ɑ	sub	sum	sob	psalm	humbly	promptly
ʌ	ɑ	putt	pun	pot	pawn	rugged	possum
ʌ	ɑ	pup	pump	pop	pomp	sponge	lobster
ʌ	ɑ	suck	sung	sock	song	substance	gossip
ʌ	ɑ	lug	lung	log	long	stomach	closet
ʌ	ɑ	hut	hunt	hot	haunt	such	nonsense
ʌ	ɑ	tuck	tongue	talk	tong	jump	problem

B F1-nasalance correlation tables

This section contains a table of F1-nasalance correlation values obtained using Spearman’s rank order correlation test ρ_s on data from Experiment 1. Confidence intervals were calculated using Fisher’s z transformation as modified by Fieller, Hartley, and Pearson (1957) with a confidence level of 95%. See 2.2.2 for details on these calculations.

Vowel	N	Oral			N	Nasal		
		ρ_s	95% CI	p -value		ρ_s	95% CI	p -value
/i/	2,261	-.19	-.23, -.15	< .0001	2,005	-.06	.01, .10	< .05
/u/	2,239	.03	-.01, .08	$\geq$.05	2,278	.13	.09, .17	< .0001
/ɪ/	4,988	.09	.07, .12	< .0001	5,049	-.01	-.04, -.01	$\geq$.05
/ɛ/	3,996	.12	.09, .15	< .0001	3,094	.07	.04, .11	< .0001
/ʌ/	3,865	.19	.16, .22	< .0001	3,842	-.08	-.11, -.04	< .0001
/æ/	2,373	.05	.01, .09	< .05	2,250	.02	-.02, .06	$\geq$.05
/ɑ/	3,570	.18	.15, .22	< .0001	2,686	-.01	-.05, .02	$\geq$.05

Table 9: F1-nasalance ρ_s values for each vowel in nasal and oral contexts.

C Experiment 2 stimuli Klatt synthesizer parameters

The following are the complete Klatt grid settings with which the stimuli used in Experiment 2 were produced. Table 10 shows settings common to all four stimuli; values which vary between stimuli are indicated with variables and are given in Table 11.

Seg	b	b	V	V	V	V	V	V	n	n	n
T	0	80	90	100	110	200	250	270	280	285	360
AV	45	45	60					60	54		54
F0	100	100		125		125			100		100
F1	200	200			x		x		480		480
F2	900	900			1050		1050		900	1270	1270
F3	2100	2100			2350		2350		2100	2130	2130
F4	2950	2950			3300		3300		2950	2990	2990
F5	3350	3350			3750		3750		3350	3400	3400
B1	500	500	80						80	40	40
B2	500	500	70						70	200	200
B3	500	500	70						70	200	200
B4	500	500	250						250	500	500
B5	700	700	250						250	700	700
NP					y		y			270	270
NZ					z		z			450	450
BNP	90										90
BNZ	90										90

Table 10: Klatt grid parameters common to all four stimuli. *Seg* indicates the segment to which certain timepoints correspond, and is included only for human readability; it is not part of the input to the Klatt synthesizer. *T* is the timepoint in milliseconds. *F* indicates formants and *B* their bandwidths; *NP* indicates the nasal formant, *NZ* the nasal antiformant, and *BNP* and *BNZ* their bandwidths.

	o	õ	u	ũ
x	430	430	415	415
y	255	255	240	240
z	265	315	250	300

Table 11: Klatt grid parameters which differ between stimuli. To produce the vowel at the top of each column, the values below it should be substituted for the variables x, y, z in Table 10.